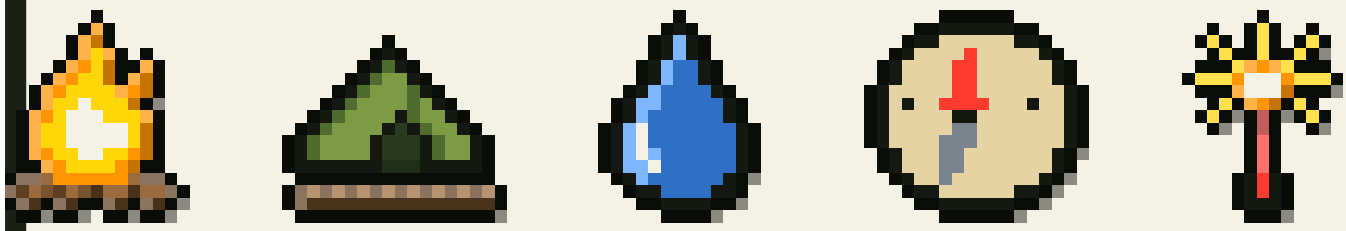
SERE PIXEL

Survival manual and training material



Printable edition · English · v4.1.0
https://strattila90-hue.github.io/SERE/

112

Practice aid — it does not replace official training and the current regulations.

Contents

Part I · Quick aids

SERE priorities 4
S-T-O-P – when in trouble 4
I'm in trouble – step by step 5
Quick cards 10
Checklists 14
Survival kit 15
Who are you? – tips by role 16
Kid mode – cards for children 18

Part II · Manual

1 The SERE training system 22
2 Fire making 24
3 Building shelters 29
4 Finding and purifying water 36
5 Evasion 39
6 Land navigation 42
7 Extraction – link-up procedures 45
8 What is expected of the person to be recovered 48
Bravo 1 Advanced fire, shelter and weather 50
Bravo 2 Water and food in the field 52
Bravo 3 Advanced navigation 54
Bravo 4 Evasion in the field, signalling, link-up 55
Charlie 1 Risk of capture and how to prevent it 57
Charlie 2 Rights and conduct of a prisoner of war 58
Charlie 3 Resilience and survival in captivity 60
Charlie 4 Escape, extraction and reintegration 61
Extra 1 Nuclear situations and doctrine 62

Part III · Practice exam questions

Exam questions 66
Answer key 79
Notes, important numbers 81

SUPPLEMENTED = from public sources; check against the current regulations.

Part I

Quick aids

At a glance: priorities, S-T-O-P, what to do when in trouble, picture cards, checklists and kit. Print it and keep it in your pack.

SERE priorities

1 

Signaling
Report the event, prepare your signalling devices. Rescue starts with a signal (Report → Locate).

2 

Protection
First aid, shelter, clothing, fire – protection from weather and enemy.

3 

Health
Treat injuries, hygiene, sleep, morale. The will to survive is your most important gear.

4 

Travel
Only with good reason and a plan: stay and wait for rescue, or exfiltrate back.

5 

Sustenance
Water first, food second. 3 days without water, 3 weeks without food.

S-T-O-P - when in trouble

S **Stop**

- Stop, take cover
- Breathe deeply, calm down
- Don't run, don't rush

T **Think**

- What happened? Who is with me?
- What injuries are there?
- What danger threatens right now?

O **Observe**

- Surroundings: terrain, weather, time of day
- Signs of enemy or locals
- What do I have? (gear inventory)

P **Plan**

- Order of priorities
- Stay or move?
- Report, signal, shelter, water

I'm in trouble - step by step

I'm lost · I don't know where I am

1. STOP. Don't keep walking blindly – every unnecessary step makes things worse.
2. Calm down: sit down, take a sip, take a few deep breaths (S-T-O-P).
3. Think it through: where did you come from, and where did you last see a known point?
4. Check your position: GPS coordinates / MGRS in the Map marker.
5. Orient yourself: compass, sun, terrain features (ridge, stream, road).
6. If you have signal: report (who, where, condition). If not: stay put if someone expects you.
7. If you must move: hold one direction, mark waypoints, go downhill: stream → road → settlement.
8. 1–2 hours before dark: shelter and fire, not marching.

I must spend the night here · shelter + fire

1. Start now – at least 1–2 hours before dark.
2. Site: not in a stream bed, not in a valley bottom, no dead branches above you.
3. Insulate from the ground: leaves, pine boughs, foam pad – this matters more than the roof.
4. Roof: tarp/poncho at 45–60°, entrance at 90° to the wind.
5. Fire: tinder, kindling, fuel wood – 3× as much as you think.
6. Fire about 1 m from your bed; fire guard, extinguishing material at hand.
7. Dry clothes for the night, dry the wet ones. Eat and drink before sleeping.

I have no water · find + purify

1. Don't waste it: shade, rest in the heat, talk little, breathe through your nose.
2. Search: valley bottoms, green vegetation, game trails downhill, birds at dawn.
3. Collect: rain on a tarp, dew with a cloth, transpiration bag on a living branch.
4. Filter: cloth → sand → charcoal → gravel. Filtering does NOT disinfect!
5. Purify: boil 1 minute, or tablet + 30 minutes (4 hours when cold).
6. Drink in moderation, with salt/electrolytes: 0.5–0.8 l/hour under exertion.
7. NEVER seawater, urine or alcohol.

I'm injured · first aid

1. Safety: cover, remove the danger.
2. Massive bleeding: tourniquet high and tight, time written down; on the torso pack the wound + pressure.
3. Airway and breathing: chin lift, recovery position, seal chest wounds.
4. Shock: lay them down, legs up, cover them.
5. Against hypothermia: insulate below and above, emergency blanket, wind shelter.
6. Report/signal as soon as you can. Moving while injured is a last resort.

Enemy nearby · evasion

1	Leave the landing / separation site immediately – that's where they look first.	
2	Hide: low, irregular, blends into the terrain – not the obvious bush.	
3	Light, noise and fire discipline. Phone screen, GPS: only if you must.	
4	Watch out: movement, shape, shadow, shine and silhouette give you away.	
5	Move at dawn, dusk or night; not on roads, not on ridgelines, not in the open.	
6	Tracks: leave no litter or broken branches; step on hard ground.	
7	Goal: designated extraction point / rally point / friendly forces.	★

Awaiting rescue · signal + link-up

1	Signalling devices out: mirror, whistle, strobe, flare, panel.	★
2	Site: an open, obstacle-free area nearby, wait concealed.	★
3	Signal only on instruction or when the rescuer approaches.	★
4	Weapon on safe, hands visible, don't run toward the rescuer.	
5	Authentication: ISOPREP questions, daily code.	
6	Accept being searched, follow instructions exactly.	
7	No radio / phone? Use signalling devices in the assigned time window at the extraction site – or exfiltrate back to friendly forces.	★

No signal / no phone · signalling without help

1. If you have a phone but weak signal: text 112 (where SMS to 112 is supported). Write WHO you are, WHERE you are (coordinates), WHAT happened. A text can get through on weaker signal than a call.
2. Go higher, to open ground (ridge, clearing, hill) – there's often signal there. Call 112 again: an emergency call can go through another carrier's network.
3. Save battery: airplane mode, switch on for 1–2 minutes each hour to try. In cold, keep the phone close to your body.
4. If someone knows your route: STAY PUT. A person who stays in one place is much easier to find.
5. Signal in THREES – the international distress signal: 3 whistle blasts, 3 shouts, 3 fires in a triangle, 3 flashes.
6. Large ground signal in the open: SOS or X (need help). Contrasting material: branches, stones, clothing, stamped into snow.
7. By day: smoke (green leaves on the fire), bright clothing spread out, signal mirror toward the sun. By night: fire, flashing light.
8. If you must move: downhill → stream → road → settlement. Leave a trail: an arrow of sticks pointing your way, tape, a cairn.
9. First inhabited place, a house, a driver, railway tracks, a power line corridor – these lead to people.

Dangerous situation · threatening person, unrest

1. Don't confront: move away calmly, don't run in panic unless you must.
2. Go where there are people: a lit area, a shop, a petrol station.
3. Call 112: say where you are and what's happening. If you can't talk, leave the line open.
4. With a child: hold their hand or pick them up, don't let go.
5. If you're trapped / cut off: hide, silence your phone, lock/barricade the door.
6. Afterwards: write down what you saw (time, place, description).

Calling for help · 112 + signalling

1. Call 112. Even if your own carrier has no signal, try anyway: an emergency call can go through any network.
2. Can't talk, or the line keeps dropping? Text 112 where supported: who you are, where you are, what happened.
3. Say WHO you are, WHERE you are (coordinates in the Map marker), WHAT happened, how many people, injuries.



4. Save battery: dim the screen, close unneeded apps, airplane mode between calls.
5. Stay put if they know where to look. You're a bigger signal than you think: bright colours, movement.



6. Signal: 3 short whistle blasts, a light, a mirror, bright clothing spread out.



7. If you hear/see searchers: wave both arms, blow the whistle, shout.

Quick cards

Fire making FIRE CRAFT

1. Gather

- Tinder: birch bark, dry grass, tinder fungus
- Kindling: dry twigs as thick as a pencil
- Fuel wood: wrist-thick and up
- 3× more than you think!

⚠ Avoid green and wet wood

2. Build

- Clear the spot (approx. 1 m)
- Teepee: tinder in the middle
- Kindling around it, leaning
- Opening on the windward side

✓ Air is half the fire – leave a gap

3. Light it

- Back to the wind, crouched
- Light the tinder from below
- Add kindling gradually
- Then the fuel wood

✓ One match, one fire

4. Watch it

- Small fire: less wood, better control
- Never leave it unattended
- A fire guard is mandatory in a group
- Put out: water, soil, stir

⚠ Keep water or soil at hand

Shelter SHELTER

1. Site

- Away from gullies and stream beds
- No “widowmaker” branches
- Not in a valley bottom (cold-air sink)
- Water and firewood nearby

⚠ Start 1–2 hours before dark

2. Frame

- Strong, can hold snow
- Pitch 45–60°
- Stubs and sharp edges removed
- Entrance at 90° to the wind

✓ 60° = optimal water run-off

3. Cover

- Tarp taut, no wrinkles
- Natural material: shingled
- Bottom to top, overlapping
- Don't touch it from inside in rain

✓ You should not see through it from inside

4. Insulation

- Ground pad: the ground steals the most heat
- Dry leaves, pine boughs, foam pad
- It needs air; don't let it compress
- Sleeping bag dry and fluffed

⚠ Without insulation you'll freeze even at 0 °C

Water WATER

1. Find

- Spring > running water > still water
- Green vegetation, game trails downhill
- Rain, dew, transpiration bag
- Melt snow, don't eat it!

⚠ Seawater, urine: NEVER

2. Filter

- Let it settle, pour off the top
- Cloth → sand → charcoal → gravel
- Filtering does NOT disinfect
- Always purify afterwards

✓ Charcoal: against taste and smell

3. Purify

- Boil: 1 minute rolling
- Above 2000 m: 3 minutes
- Tablet: min. 30 minutes
- Cold/cloudy: up to 4 hours

✓ Boiling is the safest

4. Drink smart

- 0.5–0.8 litres/hour under exertion
- Water + electrolytes together
- Survival mix: ½ tsp salt / 1 l
- Urine: pale straw-yellow

⚠ Too much water: hyponatraemia!

Signaling SIGNALING

1. Prepare

- Signal mirror, whistle, strobe
- Flare, smoke signal, panel
- Radio, spare batteries
- All on your body, not in the pack

✔ Two is one, one is none

2. Location

- Open area, clearly visible
- Contrasting background
- Concealed spot close to the signal site
- In hostile territory: only on instruction

⚠ The enemy can see too!

3. Signal

- Mirror: aim between two fingers
- Whistle: 3 short = distress
- Fire: 3 fires in a triangle
- Smoke by day, light by night

✔ Three of anything is the international distress signal

4. Ground signals

- V = need assistance
- X = need medical help
- → = travelling this way
- Signs min. 3 m long, 1 m wide

✔ Contrast: dark on snow, light on grass

Navigation LAND NAV

1. Orient

- Map oriented to north
- Compass on a north grid line
- Or by terrain features
- Declination taken into account

✔ Grid line = 1 km (1:50,000)

2. Measure

- Compass-to-cheek hold, thumb in the loop
- Target on the sighting wire
- Dial settles → read it
- Repeat

⚠ Metal, vehicles, weapons deflect it

3. Move

- Bezel: 1 click = 3°
- Intermediate landmark on the bearing
- Pace count (approx. 65/100 m)
- Check regularly

✔ At night: clicks + luminous dot

4. Fix position

- Modified resection: on a known line
- Resection: 2–3 landmarks
- Back azimuth = azimuth ± 180°
- Grid: right, then up

✔ Intersection = your position

First aid FIRST AID

1. Safety

- Cover, remove the threat
- Your own safety first
- Pull the casualty out of fire
- Assess the injuries

⚠ A dead rescuer helps no one

2. M - Bleeding

- Life-threatening limb bleeding: tourniquet
- High, tight, time written down
- On the torso: packing + pressure dressing
- Pressure, pressure, pressure

✓ Massive bleeding comes first

3. A-R-C

- A: airway – chin lift, recovery position
- R: breathing – seal chest wounds
- C: circulation – signs of shock, legs up
- Continuous checks

✓ MARCH sequence

4. H - Hypothermia

- Insulate below and above
- Emergency blanket, dry clothes
- Wind shelter, warm drink if conscious
- Injured people cool down faster

⚠ Cold also worsens bleeding

Checklists

Before deployment

- [ ] ISOPREP filled in, up to date
- [ ] Evasion plan of action (EPA) known
- [ ] Comms plan: frequency, call sign, time window
- [ ] Survival kit on your body
- [ ] Signalling devices checked
- [ ] Map: extraction sites, rally points
- [ ] Water, electrolytes, purification
- [ ] Compass checked

Shelter - considerations

- [ ] Is it needed at all?
- [ ] Do I have enough energy?
- [ ] What do I have?
- [ ] How long is it for?
- [ ] Is it protected from the elements?
- [ ] Does it need camouflage?
- [ ] Is it exposed to natural hazards?
- [ ] Is heating needed?
- [ ] Is ventilation ensured?

Fire - preparation

- [ ] Area cleared (horizontally and vertically)
- [ ] Wind direction checked
- [ ] Tinder dry and plentiful
- [ ] Kindling sorted by size
- [ ] Firewood 3× your estimate
- [ ] Fire starter at hand
- [ ] Extinguishing material (water/soil) at hand
- [ ] Fire guard assigned (in a group)

Before link-up

- [ ] Designated site reached, surroundings observed
- [ ] Check in only on instruction
- [ ] Signalling devices ready
- [ ] Weapon on safe, hands visible
- [ ] ISOPREP answers memorised
- [ ] Landing zone clear of obstacles
- [ ] Follow instructions exactly

Survival kit

Core kit (always carry it)

- Knife
- Fire starters (2–3)
- Canteen + cup
- Water purification tablets
- Individual first aid kit (IFAK)
- Compass (Cammenga)
- Signal mirror
- Whistle

	Item	Priority	Weight	
	Knife	RARE	150 g	☐
	Fire starters (2–3)	IMPORTANT	80 g	☐
	Canteen + cup	BASIC	350 g	☐
	Water purification tablets	IMPORTANT	20 g	☐
	Individual first aid kit (IFAK)	IMPORTANT	400 g	☐
	Paracord 5–6 m	BASIC	60 g	☐
	Compass (Cammenga)	RARE	200 g	☐
	Map + protractor	RARE	120 g	☐
	Tarp / poncho	IMPORTANT	700 g	☐
	Emergency blanket	BASIC	60 g	☐
	Signal mirror	IMPORTANT	40 g	☐
	Whistle	BASIC	15 g	☐
	Signal light / flare	RARE	150 g	☐
	Electrolytes + salt tablets	IMPORTANT	50 g	☐
	Energy bars, dried fruit	BASIC	300 g	☐
	Axe / saw	RARE	900 g	☐
			3595 g	

Who are you? - tips by role

Soldier

land forces, SERE level A

You're alone, separated from your unit. The goal: survive, stay undetected, get back or await extraction.

- ISOPREP, EPA, comms plan memorised
- Survival kit on your body, not in the pack
- Light, noise and fire discipline
- Captivity is worse than evasion

Parent with young children

mum / dad with kids

Children get cold faster, dehydrate faster and tire sooner. Their pace and condition decide.

- Keeping the child warm and dry comes first
- Give them a drink every hour, small sips – don't wait until they ask
- Teach them: if lost, HUG A TREE and stay there
- Give them a whistle – 3 blasts = help
- Make tasks a game: “let's collect sticks for the fire”
- Calm voice: they read the situation from your face

Pregnant / with a baby

pregnancy, breastfeeding

You and the baby need more fluids, you tire more easily, heat and cold hit you harder. Avoid heavy work.

- More water: drink every hour
- Don't lift heavy things – ask for help
- Boiling or chlorine dioxide instead of iodine
- Breastfeeding gives the baby both fluids and warmth
- Abdominal pain, bleeding, reduced baby movements: 112 immediately

City dweller

flat, public transport

In a city, water, power and crowds are the key. If services stop, the shops empty within days.

- 3 days of water at home: 3 litres/person/day
- Charged power bank, torch, battery radio
- Cash in small notes – cards don't always work
- A family meeting point: where you meet if there's no signal
- Know your stairwell and emergency exits
- Avoid crowds and panic buying

Rural / village dweller

garden, forest, help far away

In the countryside help is farther away and slower to arrive, but there are more natural resources: wells, wood, gardens.

- The ambulance takes time: you'll have to manage the first hours yourself
- Well water: boil it after floods or contamination
- Spare firewood, candles, oil lamp
- Axe, saw, rope at hand
- Know your neighbours – you depend on each other
- Animals need water and feed too

Hiker / day-tripper

forest, hills, weekend

Most hiking accidents come from bad timing, wrong clothing and too little water.

- Always tell someone where you're going and when you'll be back
- Offline map, charged phone, power bank
- Headlamp – even if you plan an afternoon trip
- Layered clothing, rain jacket
- Turn back in time: darkness won't wait

Elderly / limited mobility

medication, slower pace

Medication, warmth and water matter most. Don't overdo it – calling for help is not weakness.

- Medication for at least 7 days, list on paper
- Spare glasses, hearing aid batteries
- You cool down faster in the cold – dress in layers
- Don't wait long to call 112
- Make sure a neighbour/relative knows about you

For women - additions

- Spare period products (pads, tampons or a menstrual cup)
- Cup/tampon only with clean hands – sanitiser gel, purified water
- Keep dry and clean: infections spread fast in the field
- When alone: tell someone your route; don't give strangers your exact location

Kid mode - cards for children

Read it aloud or talk it through with your child. Cut it out and put it in their bag.



I'm lost

1. It's okay that you got lost. Now listen to me.
2. Stop. Don't walk any further.
3. Find a big tree and hug it. Stay there by the tree.
4. Blow your whistle three times. If you have no whistle, shout: Help!
5. The grown-ups will come looking for you. You just stay by the tree.
6. If you have a phone, call one-one-two.



I'm scared

1. You're brave. Let's breathe together now.
2. Breathe in slowly: one, two, three, four.
3. Breathe out slowly: one, two, three, four.
4. Once more. You're doing really well.
5. If you're in trouble, call one-one-two, or tell a grown-up who can help: a police officer, a shop worker, a mum with kids.



I'm injured

1. If it's bleeding, press hard on it with your hand or a cloth, and don't let go.
2. Sit or lie down.
3. Call one-one-two, or tell a grown-up.
4. If a grown-up is hurt and not talking: call one-one-two and say what you see.



I'm cold

1. Put on everything you have with you. Your hat is very important.
2. Don't sit on the bare ground. Sit on your bag or on leaves.
3. Get out of the wind: behind a tree, next to a bush.
4. Curl up small and tuck your hands under your arms.
5. If your clothes are wet and you have dry ones, change.



I'm thirsty

1. Drink the water you brought from home.
2. Don't drink from streams or puddles – it can make you ill.
3. Stay in the shade and don't run around.
4. Tell a grown-up that you're thirsty.



I have no phone

1. It's okay, they'll find you even without a phone.
2. Stay where you are. Hug a tree, or sit somewhere easy to see.
3. Blow your whistle three times. If you have no whistle, shout three times: Help!
4. If you hear someone calling you, shout back and wave both arms.
5. If you have brightly coloured clothes, put them where they're easy to see.
6. If you see a grown-up who is working or helping, go and tell them.



I can't find mum or dad

1. Stay where you are. Mum and dad will come back for you.
2. If you're in a shop, tell the cashier or a security guard.
3. On the street, tell a police officer, or a mum who has kids with her.
4. Don't go anywhere with a stranger.
5. Say your name and, if you know it, mum's or dad's phone number.



A stranger wants to take me

1. If a stranger tells you to go with them, say it loudly: NO! I am not going!
2. Don't get into anyone's car. Don't take sweets or presents from them, and don't go with them to see a puppy.
3. Don't tell them where you live, and don't tell them you are alone.
4. If they say Mum or Dad sent them, don't believe it. Only go with someone Mum or Dad told you about before. If your family has a secret word, ask for it. If they don't know it, don't go.
5. Run to where there are lots of people: into a shop, to a mum with children, or to a police officer.
6. If they grab you, shout very loudly: Help! This is not my dad! Wriggle, kick, and run away.
7. Tell Mum or Dad what happened. You are not in trouble. You did really well to tell them.



Someone took me and I am scared

1. You are brave. Breathe in slowly, and breathe out slowly.
2. If you have a phone, call one-one-two as soon as you can. If you can't talk, don't hang up, keep the line open.
3. If there are people nearby, shout loudly: Help! This is not my mum! This is not my dad!
4. Notice what you see and hear: what the car looks like, what the house looks like, what sounds there are.
5. If you can get away, run to where there are people, and tell a police officer or a grown-up who is working there.
6. If they say Mum and Dad are not looking for you, don't believe it. Mum and Dad are always looking for you.
7. Whatever happens, it is not your fault. They will find you.



112

Part II

Manual

The full course: SERE level “A”, “B”, “C” and the EXTRA chapter.

1 The SERE training system

Part I, Lesson 1

1.1 Introduction

Our world is becoming ever more civilised, and in the process the instincts needed for survival are fading. To protect soldiers, special attention must be paid to restoring and developing their ability to survive.

SERE is an English acronym: **S**urvival, **E**vasion, **R**esistance, **E**xtraction.

The training covers tactics, techniques and procedures that reduce the chance of an isolated person being captured and significantly improve their chances of staying alive in different environments. If captured, they can resist and survive captivity and, if the situation allows, escape.

1.2 About this course

The unified SERE training system ensures that everyone knows the SERE rules in the detail required for their level – "A", "B" or "C" – so that the risk of losing or having personnel captured is reduced if they become isolated.

> **Theory is essential, but it never replaces practice. Hear it and you forget; see it and you know it; do it and you can do it!**

- Low risk category → SERE "A"
- Medium risk category → SERE "B"
- High risk category → SERE "C"

1.3 SERE training in general

In the Hungarian Defence Forces, the SERE training system is regulated by a directive of the Chief of Defence.

1.4 Key terms

- **ISOP – Isolated Personnel:** a soldier or civilian separated from their unit or organisation who has lost positive and/or procedural control and must use survival, evasion and resistance techniques in order to return.
- **Survival:** all activities that allow the isolated person to stay alive and maintain physical and mental health under various environmental conditions.
- **Evasion:** procedures that allow the isolated person to avoid detection and contact by hostile or potentially hostile forces and people, or to enable return.
- **Resistance:** tactics, techniques and procedures that allow the isolated person, if captured, to reduce the effectiveness of the captors' interrogation tactics.
- **Extraction:** recovering the isolated person from hostile or potentially hostile territory.
- **JPR – Joint Personnel Recovery:** the sum of military, diplomatic and civil efforts to recover and reintegrate isolated personnel.

1.5 Principles

Recent operations show a significant chance that both soldiers and civilians become isolated, are captured and face coercive interrogation. Participants must be able to survive, evade the enemy, prepare for extraction or exfiltrate back to friendly forces; if necessary resist coercive interrogation and assist in their own recovery, including escape from captivity.

Because modern battlefields are vast and technology advances, our troops operate at ever greater distances, which increases the chance of isolation. ISOP must be prepared for recovery and self-recovery, trained, and equipped with the necessary gear.

Losing soldiers trained on advanced equipment can mean a decisive loss of capability, so they must be prepared to survive, to exfiltrate (escape) or to prepare for extraction.

Most theatres have a JPR system, whose effectiveness depends not only on equipment but also on how well prepared the person to be recovered is.

1.6 The SERE training system

Training can be theoretical or practical depending on the level, environment, threat and theatre. Training should be carried out with exactly the equipment that will be used on operations. Success depends on uniform training, preparation and proper equipment – their absence can even be a strategic risk.

Anything not in line with the valid SERE training programme cannot be considered SERE training. The programme is prepared by the permanent SERE group and approved by the training directorate.

Nationally the system follows NATO STANAG 7196 (ED 2) – SERE Training Standard, APRP-3.3.7.5 Edition B.

1.7 Levels of the training programme

- **SERE “A”:** theoretical training for personnel at low risk of isolation; delivered by the units' training staff.
- **SERE “B”:** theoretical and practical training for medium-risk personnel; delivered by the central SERE training group with the units.
- **SERE “C”:** training for high-risk personnel; coordinated by the subject-matter lead and delivered by the central SERE training group.

> ✔ Levels “B” and “C” are valid for 5 years and can be extended by 3 years with refresher training – any number of times.

1.8 The SERE “A” course

SERE “A” is theoretical training for personnel at low risk of isolation. Each chapter takes about 1–2 hours and ends with a 10-question test; the pass mark is 70%.



2 Fire making

Part I, Lesson 2 – Lighting fires with lighters, matches and improvised means

2.1 Why fire matters

Being able to make fire is one of the most important survival skills. Fire can be used:

- to warm yourself, heat things up and dry clothes;
- as a light source and for signalling (light, flame, smoke);
- for making tools, cooking and baking;
- for purifying water (boiling, charcoal);
- to scare off wild animals and keep insects away (smoke);
- to lift morale, and charcoal for camouflage.

✔ When you use fire to warm up, your body burns fewer calories to keep its temperature – you need less food.

⚠ In hostile territory the light, smoke and smell of a fire can easily give away your position. Knowing the methods is not enough – you need skill gained through practice.

2.2 Considerations

- Avoid big fires: a small fire needs less fuel, is easier to control and its heat is more concentrated.
- Never leave a fire unattended unless you have safely banked or put it out.

✔ Banking: rake cold ash and dry soil over the fire, leaving enough air at the top for the fuel to smoulder. It can be revived later.

2.3 Safety

Distinguish between an emergency and peacetime training bound by rules. Always keep in mind the protection of human life, animals, nature, property and facilities.

⚠ Never underestimate the power of fire – it can get out of control in moments.

⚠ In cold weather burns are more common than cold injuries. Use protective gloves, a multitool, an insulated handle.

⚠ Exhaustion and carelessness can burn equipment too (radio, boots, poncho) – fatal in a survival situation.

⚠ In enclosed or poorly ventilated spaces smoke (carbon monoxide) poisoning can occur – ensure ventilation.

- In group or pair training, always assign a fire guard.
- Keep extinguishing means (water, sand, spade, shovel) at hand.

2.4 Elements of fire

The three basic elements of a successful fire are the **fire triangle**:

1. fuel;
2. heat;
3. oxygen.

Limiting the fuel only allows a small fire. If the fire is not fed properly, it can be too big or too small. Freshly cut green wood ignites poorly – the fire must already be burning well to use it.

> ✔ Taking away any element (covering, water, removing fuel) puts the fire out – think this way when extinguishing.

2.5 The three levels of fuel

1. **Tinder** – easily ignited material;
2. **Kindling**;
3. **Fuel** – firewood.

> ★ **When you think you've got enough tinder, kindling and firewood, triple it – and then maybe it'll be enough.**

2.6 Tinder

Any small material with a low ignition point that catches from minimal heat (e.g. a spark). Arrange it so that air can get between the hair-thin, bone-dry fibres.

> ✔ Dry tinder is vital for the isolated person – always keep some at hand in a waterproof pouch.

- It can be collected in two or three stages: keep collecting and drying it as you move (e.g. in your pocket, close to your body).

2.7 Natural tinder

- shredded bark of certain trees and shrubs: cedar, palm fibre, **birch bark** (burns even when wet);
- the lining of bird or rodent nests;
- charred rotten wood (punk wood);
- dry tinder fungus;
- the fluff of milkweed, thistle, clematis, reed;
- crushed fibres of dead plants (straw, hay);
- fine, dry wood shavings;
- scrapings of resinous wood (fatwood);
- dried, powdered conifer resin (pitch).

2.8 Artificial tinder

- cotton wool and products: tampons, cotton pads;
- char cloth, cotton balls, lint;

- plastic shavings (e.g. an MRE spoon);
- petroleum fuels (kerosene, petrol);
- steel wool (also lights from battery terminals);
- paper, foam;
- alcohol-based hand sanitiser;
- gunpowder;
- fire-starter tablets.

> ⚠ Petrol vapour is explosive – use only a small amount, pour it on cold tinder before lighting.

2.9 Kindling

The next level of fuel. It has a higher ignition point and must be placed so it catches when the tinder's flame reaches it. Its job is to raise the burning temperature to where larger fuel can feed the fire.

Natural: dead, dry small twigs; thinly split pieces of wood, bamboo or reed; shavings carved from the inside of larger branches; cones, needles; the dead lower branches of conifers.

> ⚠ Always split bamboo and elder – whole, they can explode because of the trapped air.

Artificial: some plastics (MRE spoon); wood soaked in wax, oil, insect repellent; vaseline-soaked gauze from the first aid kit.

> ✔ Split wood burns more easily because it's drier inside; sharp edges catch more easily. Sorted fuel catches more easily.

2.10 Fuel

Firewood doesn't need to be completely dry – a steadily burning fire dries it. Still, protect it from moisture because of smoke.

> ⚠ Never pour flammable liquid onto a burning fire – even a smouldering fire can flare explosively.

- Dry split hardwood (oak, hickory, ash): little smoke, lots of heat, but harder to cut.
- Pine and conifers: burn fast, smoke a lot.
- Rotten wood: little heat, mostly smoulders and smokes.
- Standing or leaning dead trees: usually dry inside even after rain.

Recommended fuel: standing dead trees and dry branches that snap; the inside of fallen trees; as a last resort green wood split fine and mixed with dry; in treeless areas twisted bundles of dry grass, cactus, dry peat moss, dry animal dung, animal fat, oil-soaked sand.

2.11 Preparing to make fire

Take enough time to prepare: collect every level of fuel and prepare the fire starter **before** lighting.

> ✔ Success = preparation + practice + patience.

2.12 The fire site

Clear the fire site carefully and keep a safe distance from natural (leaf litter, dry grass) and man-made (pack, tent) flammables – horizontally and **vertically** too (washing line, low branches).

> ⚠ Wind strength and direction can change – embers can burn the shelter tarp or start a fire.

- To contain the fire: metal stove, a pit dug in the ground, a ring of stones.

> ⚠ Don't dig a fire pit near pine trees: resinous roots act like wicks and can start a forest fire days later.

> ⚠ Wet, porous or hollow stones can explode from the heat – don't use riverbank stones.

- In snow, dig down to the ground – the pit must be wide and airy enough, or melting snow or lack of oxygen will put it out.
- On wet, marshy ground you need a platform (stones, green logs).

2.13 Get the most heat

Use a **heat reflector** built of stone, wood or other material (emergency blanket, metal sheet) to direct heat into the shelter. Logs/stones around the fire support a cooking pot and also act as a windbreak.

2.14 Rationing fuel

Stack and sort it by size (tinder, kindling, firewood).

> ★ **Rule of thumb: collect three times as much tinder and kindling as seems necessary.**

> ✓ For a long stay, ration responsibly and protect the environment – don't waste!

2.15 Building a fire platform

You need a base if the fire would melt through deep snow, or the ground is wet or marshy. It can be made of green logs, metal or non-flammable material.

> ⚠ On humus or peat soil you always need a platform: a smouldering peat fire is almost impossible to put out and can burn for years.

2.16 Fire starters

Should be quick and easy to use, even with gloves: matches (storm matches), lighter, storm lighter, magnesium block, ferro rod, flint striker.

> ✓ Always carry at least 2–3 independent fire starters stored in separate places (“two is one, one is none”).

2.17 Arranging the kindling

Arrange a small amount of kindling in a low teepee, close enough for the flames to jump across. Leave a small opening for lighting and airflow. Paper is the easiest tinder.

2.18 Using a feather stick

You can save matches with a feather stick or an airy bundle of thin dry twigs. Shield the flame from wind; watch the wind direction so the flame doesn't catch your hair, clothes or gear. Place the tinder under the kindling on the windward side.

2.19 Small sticks to support the kindling

Before or after lighting, carefully add small sticks to the kindling, then gradually bigger ones. A forearm-thick piece of wood as a support stops the fire from being smothered and keeps it ventilated.

2.20 Ways to light a fire

> **Teaching and using these is part of SERE "B" and "C" training – here they appear as knowledge only.**

Sunlight (magnification) – needs strong, clear sun and good tinder: magnifying glass (also on a compass), glasses, Fresnel lens, magnifying mirror, the reflector of a torch, a polished drinks-can bottom, a crystal-clear bottle/glass/bag of water.

Electricity – heat of a short circuit: thin wire, chewing-gum foil, steel wool + battery.

> Electrical methods carry a risk of accidents and explosion and can cause a fatal electric shock!

Friction – bow drill, hand drill, fire plough, bamboo fire saw, fire roll.

Spark – ferro rod; flint and steel (knife, file).

2.21 Other methods

Mini stove (solid-fuel stove): compact folding steel stove with solid fuel tablets. One tablet is 10–15 g and burns about 8–10 minutes – for heating food without open fire.

Plastic items: stick the handle of a plastic spoon upright in the ground and light the tip – it burns about 10 minutes, enough to dry and light tinder. A candle saves matches the same way.

2.22 Fire lays

Most fires are built for a specific need (warmth, light, boiling water, cooking). Names vary in the literature.

- **Teepee:** good light, concentrated heat above the top, ideal for boiling water. Tinder in the middle, kindling leaning on a stake around it, opening on the windward side. Light it with your back to the wind.
- **Log cabin:** even, long-burning bed of embers, for cooking.
- **Long fire (Finnish fire):** two logs on top of each other, burns all night, for warmth.
- **Upside-down (top-lit) fire:** needs little feeding.
- **Star (shepherd's) fire:** burns economically as you push the logs inward.
- **Lean-to fire:** sheltered from wind, kindling leaning.
- **Dakota fire hole:** two connected pits – little light and smoke, burns well in wind, good for concealment.



3 Building shelters

Part I, Lesson 3

3.1 Introduction

At SERE “A” level every soldier is expected to build, on their own and from issued equipment, a shelter that protects against the harmful effects of nature in a friendly environment under field conditions. E-learning alone is not enough – you have to practise.

3.2 Shelter

A shelter can be anything that protects you from environmental hazards. The environment and the situation affect the choice of site and the factors the isolated person must consider.

3.3 Shelter Factors

When choosing a site, consider:

- the time and energy needed to build it;
- how long you'll stay (short or longer);
- the weather (temperature, wind, sun, precipitation);
- life forms (human, plant, animal);
- the terrain;
- the time of day;
- the situation (hostile or neutral environment).

> ✔ Maximum protection for minimum energy!

3.4 Considerations - checklist

- Is it needed at all?
- Do I have enough energy to build it?
- What do I have to build it with?
- How long is it for?
- Is it protected enough from the elements?
- Does it need camouflage?
- Are there objective hazards threatening it?
- Do I need heating?
- Is proper ventilation ensured?

> ⚠ Always allow enough time to choose the shelter site!

3.5 Timing

Don't wait until the last moment: in bad conditions you may be forced to use poor materials and a poor site. Set a deliberate order of priorities between fire, water, food and shelter.

✔ Rule of thumb: start building at least 1–2 hours before dark.

3.6 Weather and temperature

Ignoring the weather can have disastrous consequences.

- Valley bottoms are colder – a **"cold-air sink"**: cold air flows down here at night.
- In cold months choose open, sunny ground; in hot months shaded, covered ground.
- In deserts it's scorching by day and freezing at night – you must protect against both.

3.7 Wind

In summer a cool breeze helps and keeps insects away (hilltop, island). Blowing sand, dust or snow irritates skin and eyes and damages equipment; take extra care with fire. In cold you need protection against wind and drifting snow.

3.8 Precipitation

⚠ Don't build in a gully, drainage channel or dry stream bed – sudden floods, mudslides, flash floods.

⚠ In snowy mountains avoid avalanche-prone slopes.

3.9 Life forms

Consider every life form. Humans may be the enemy or another group you must remain unseen by.

- **Insects:** fewer on hills, ridges and breezy spots. Avoid still water (mosquitoes, bees, wasps) and anthills.
- **Animals:** don't build near game trails, dens or watering holes.
- **Dangerous plants:** thorns, vines, poisonous plants. Avoid dead, half-fallen trees and branches – **"widowmakers"** can fall in wind or rain and kill.
- With fire, fallen leaves and dense dry vegetation easily catch.

3.10 Terrain

⚠ Avoid avalanche, rockfall and landslide areas – recognisable as a clear strip of different vegetation from the top of a hill to the bottom.

⚠ Check rock shelters and overhangs before use.

3.11 Requirements and planning principles

A good shelter gives physical and mental well-being for rest, which is vital for good decisions – the longer rescue takes, the more critical this is.

- It should be adequate with little work and energy; don't limit yourself to a predetermined form.
- Use natural shelter (rock overhang, rock niche, cave, crevice, fallen log, root wall, snow wall) and improve it.
- Insulation can be increased with vegetation or a parachute; a heat reflector can be built under a ledge or in front of a cave.

3.12 Defining the shelter's purpose

The type depends on: climate; weather; insects; available materials; length of stay; the enemy's presence, numbers and equipment; the isolated person's physical condition.

> ⚠ In enemy territory think about traces while building and dismantling: minimal intervention, careful removal.

3.13 Shelter specifications

- Big enough for comfort, but not so big it's a lot of work (in cold, smaller = warmer).
- If you light a fire, ventilate for smoke and carbon monoxide – even if the fire doesn't smoke (fuel tablets, gas, hand sanitiser).
- Fire only outside the shelter; the opening at **90°** to the prevailing wind.

> ✔ Ideal fire-to-shelter distance: approx. **90–100 cm**.

> ⚠ Don't build a fire near aircraft or vehicle wreckage (spilled fuel). Move your materials a safe distance away instead.

3.14 Protection from rain and snow

With porous (non-waterproof) material use the **"pitch and tautness"** principle: a pitch of **45–60°**, with the material taut and wrinkle-free. Don't touch the material while water runs over it – it will drip through there.

3.15 Materials, site, frame

Collecting materials: frame, covering, bed/insulation, fasteners – all at hand before you start.

1. Remove all rough edges and stubs from the poles of a wooden frame.
2. For a natural shelter leave branches on the outer side of the tree – they're good supports.
3. For frame and cover: bark, cut branches, palm leaves, grass, woven vegetation.
4. With a parachute, remove the lines and cut to size.

Preparing the site: clear rocks and twigs from the bed, cut back overhanging vegetation.

Frame: it must carry snow, rain and gusts. A **60°** pitch is optimal. With a parachute, tension the centre seam, work from back to front, pull at 90° to the folds. With natural covering use the **shingle method**: from the bottom up, overlapping – you shouldn't be able to see through from inside.

3.16 Types of shelter

By material: natural, man-made, combined.

By camouflage: conspicuous or concealed.

By construction: existing, hasty, improvised.

Climate zones: hot (equatorial, tropical, desert, savanna, monsoon), temperate (oceanic, Mediterranean, continental), cold (subarctic), and mountain.

3.17 Requirements for a concealed shelter

- Close to water, food, fuel, signal or extraction site.
- Safe, protects from environmental hazards; enough material nearby.
- Big enough to sit, lie and store gear; level floor.
- Hard to approach, in the least obvious place.
- Several approach and escape routes; can be left immediately.
- Natural camouflage (against aerial observation too); irregular shape, blending colour.
- Easy to dismantle (quick-release knots).

Conspicuous (uncamouflaged) shelter: easy to approach, in an obvious place, marked with bright colour and light.

3.18 Existing, hasty, improvised

- **Existing:** ledges, fallen trunks, caves, tree wells; in towns, buildings. If one exists, use it – only build new if necessary.
- **Hasty:** quick, minimal effort: life raft, vehicle or aircraft parts, parachute.
- **Improvised:** quick to build and dismantle (lean-to, diamond, A-frame, sunshade, teepee) – only if you don't need to flee immediately.

3.19 Temperate-zone shelters

1. **Cave** – easy to heat, cold without fire; obvious place marked on maps; no escape route; predators.
2. **Rock niche, overhang** – shaded, draughty, protects from precipitation; hard to reach, no escape route.
3. **Sod shelter** – sod blocks about 5 cm thick on a strong frame; can be heated, cool in summer, water- and insect-proof, fireproof. Needs a spade; time-, material- and energy-intensive – long-term only.
4. **Debris (A-frame) shelter** – in deciduous forest, camouflages excellently; lots of material and time; insects; risky to heat with fire.
5. **Fallen tree, improved** – check the tree's stability; camouflages well, limited protection from wind and rain.

3.20 Tarp-based shelters

A soldier's own tarp-type gear: issue tent sheet, poncho, parachute, 2–3×3 m tarp, emergency blanket, bin bag.

1. **Lean-to** – the quickest, keeps rain off well, can't hold snow; windproof from one side; fire warms a large surface. At 60°+ even non-waterproof material sheds water.
2. **Modified lean-to with porch** – from an A-frame by raising one side; more cordage; take extra care with fire distance.
3. **Lean-to with ground sheet** – the folded-back part insulates against ground moisture.
4. **Diamond** – one anchor point is enough; windproof over 180°; heats well.
5. **A-frame** – excellent rain protection, windproof from two sides, hard to heat.
6. **Modified A-frame** – closed back, windproof from several sides.
7. **Raised A-frame** – hot climates (jungle); airy, no wind protection.
8. **Teepee (parachute)** – excellent against wind, rain, cold and insects, fire possible inside; too big for a concealed shelter, but good for signalling at night.

9. **Teepee (tent sheets) – "pup tent"** – two buttoned issue tent sheets; the centre pole must be stable; leaks if not taut.
10. **Burrow / bivvy** – dry insulation in a waterproof cover, "do as the animals do"; warmed by body heat.
11. **Vehicle wreckage** – wing, tail or fuselage for shade; inside overheats in heat and cools in frost; fire danger!
12. **Buildings, ruins** – abandoned buildings, sheds, garages, culverts; don't hide in the only obvious spot; change locations if possible.

3.21 Hot-zone shelters

Tropics, jungle: on hills, high ground, away from swamps; few places to build directly on the ground.

1. **Raised platform bed** – for wet, swampy ground; time-, tool- and material-intensive.
2. **Hammock** – tarp or parachute between two load-bearing points; quick to strike, leaves few traces; hard to insulate in cold.

Desert: build in the early morning, evening or at night. Shade of rocks and dunes; the material 30–40 cm above you.

1. **Improvised sunshade** – 2–3 layers with a 30–40 cm air gap between; the bed about 40 cm above or below the surface.

3.22 Snow and ice - principles

In a small, well-built snow shelter the temperature rarely drops below –10 °C. Body heat raises it by about 4 °C, one candle by another 4 °C or so.

> ✔ If you can't see your breath, it's too warm inside – cool it, or it will melt and drip.

- Three levels: fire/vent at the top, sleeping platform in the middle, cold sink at the bottom.
- Scrape off the glazed (iced) inner layer.
- Keep the old shelter until the new one is ready.

> ⚠ Physical work makes your body produce up to 1500 W of heat – take off layers and open zips before building. **Sweat is deadly in the cold!** Work slowly, at an even pace.

3.23 Entrance, ventilation, hygiene

- Opening at 90° to the prevailing wind; close the entrance with a snow block or pack.
- Take the shovel inside at night – if snow buries you, you can dig out.

> ⚠ Candles, burners and oil lamps produce carbon monoxide: have vents in the roof and by the door.

- Hygiene: stay inside if possible; a separate tunnel as a toilet; bottles, snow blocks.

3.24 Barren and wooded areas

Barren: seashore, ice cap, above the treeline; little material. Build on the lee side of wind-blown ridges. Avoid unstable ice.

1. **Snow cave** – needs a shovel; quick, excellent wind protection; ventilation; avalanche risk.

2. **Quinzee** (packed snow mound) – time-consuming, lots of energy; clothes get soaked – dry them straight away.
3. **Igloo** – needs a cutting tool and firm snow; lots of time and precision.

Wooded: plenty of material; not on riverbanks (floods, mudflows).

1. **Tree well** – the hollow under a spruce's low branches; dig from the lee side; don't knock snow off the branches.
2. **Snow trench** – at least 1–1.5 m of snow; a digging tool; quick and low-effort with a tarp; covered with boughs and snow it insulates and camouflages.

3.25 Maintenance and improvements

For longer use maintain it: a taut tarp protects better; a door against wind and insects; heat reflector, porch, work area, separate storage for gear and fuel.

3.26 Sleep system - sleeping bag

A sleeping bag only gives maximum warmth when clean, dry and fluffed. Turn it inside out, shake it, dry it in the sun or at a safe distance from the fire.

> ⚠ Don't breathe into the bag (keep nose and mouth outside) – breath makes it damp. Don't keep a water bottle in it.

> ✔ Before getting in: light exercise (without sweating), food and a hot drink.

Improvised sleeping bag: parachute, sheet or blanket folded lengthwise, sewn or tied along the bottom and side, 15–25 cm longer than your height. With a double layer fill the gap with dry grass, straw, cardboard, newspaper, foam.

3.27 Ground pad - ground insulation

You need insulation toward the ground too: natural (dry leaves, pine boughs, reed) or man-made (foam, inflatable mat). The air layer insulates – don't let it compress.

> ⚠ Even in a winter bag rated –25 °C you'll be cold at 0 °C without ground insulation. Cooling can start below 15 °C!

- **R-value:** a mat's insulating ability – the higher, the better. About 1.0 (minimal) to 5+ (excellent, winter).
- Every extra kilo counts on a march.

Man-made: sleeping bag, blanket, poncho, tent sheet, 3×3 m tarp, parachute, emergency blanket, bin bag, foam and inflatable mats.

Natural: dry grass, leaf litter, reed bundles, straw bales, pine boughs, framed bed lined with natural material.

Accessories: tent pegs, wooden stakes, 4–5 m of 5–6 mm rope, 5–6 m paracord.

3.28 Knots

> ⚠ Some knots are also used in climbing and rope work, but different rules apply there. Using them incorrectly is dangerous and can kill!

- **Overhand knot** – fastening, grip knot, rope end. + Quick, basis of other knots. − Hard to untie, can slip. ⏱ 10 s.
- **Slip knot** – running knot. Fastening. + Quick to tie and untie. − Less secure. ⏱ 15 s.
- **Overhand loop** – a fixed loop along or at the end of the rope; grip, hanging point, tensioning eye. + Can be tied when neither end is free. − Almost impossible to untie after loading. ⏱ 15 s.
- **Half hitch** – tying bundles and packages. + Easy to untie. − Insecure. ⏱ 20 s.

4 Finding and purifying water

Part I, Lesson 4

4.1 Introduction

Water is one of the most important things for sustaining human life. About **70%** of the body is water; it is essential for circulation, kidney function, breathing, digestion and temperature regulation.

> **You can survive weeks without food, but on average only 3 days without water.**

Clean drinking water is not guaranteed in a survival situation – every soldier must know how to obtain, purify and safely store water. Lack of water causes dehydration, confusion, muscle weakness and ultimately death.

4.2 Dehydration

Loss of fluids, which also involves loss of electrolytes. Three main types:

1. **Isotonic:** water and electrolytes are lost in the same proportion. Causes: diarrhoea, vomiting, burns, sweating. Signs: fatigue, flushed skin, headache, dark urine, cramps. Treatment: isotonic fluid replacement.
2. **Hypotonic:** electrolyte deficiency from replacing water without minerals. Sodium deficiency → disturbed nerve conduction, muscle cramps. Signs: weakness, nausea, oedema. Treatment: replace salts and electrolytes, restrict water.
3. **Hypertonic:** lack of fluid – the plasma thickens and draws water from the cells. Signs: strong thirst, dry mucous membranes, confusion. Treatment: gradual, slow rehydration.

> Watch your urine: pale straw-yellow = fine; dark and little = drink!

4.3 Electrolytes

Minerals that split into ions in water – key to cell function and nerve-muscle signalling. Under heavy exertion, extreme temperatures and little water, keeping them balanced is vital.

- **Sodium (Na^+):** blood pressure, blood volume, muscle and nerve function; lost quickly in sweat. Sources: salt, salty food, electrolyte drinks.
- **Potassium (K^+):** heart rhythm, muscle function. Sources: bananas, dried fruit, field rations.
- **Calcium (Ca^{2+}):** muscle contraction, bones, blood clotting. Sources: milk powder, tinned food, supplements.
- **Magnesium (Mg^{2+}):** nerve-muscle function, prevents cramps. Sources: seeds, grains.
- **Chloride (Cl^-):** acid–base balance, digestion. Source: table salt.
- **Bicarbonate (HCO_3^-):** blood pH. Made by the body, can be replaced with solutions.

Causes of loss: heavy sweating (heat, heavy kit), diarrhoea, vomiting (contaminated water), injuries, burns, prolonged lack of water.

Signs of deficiency: muscle cramps, weakness, dizziness; irregular heartbeat, low blood pressure; poor concentration; in severe cases loss of consciousness.

4.4 Replacing electrolytes

- Replace water and electrolytes **together** – drinking only water while sweating can cause hyponatraemia.
- Isotonic drink powders in MREs are absorbed quickly.
- Salt tablets: in extreme heat, in the desert.
- Survival mix: **½ teaspoon of salt / 1 litre of water**.
- Dried fruit, nuts, dried meat, energy bars – potassium, magnesium.

Mandatory items of a military survival kit: electrolyte drink powder, salt tablets, water purification tablets, compact water purifier.

4.5 Water intoxication (hyponatraemia)

A life-threatening condition: too much water in a short time while the blood's sodium level drops – cells (including brain cells) swell, causing brain oedema and even death.

Causes: the kidneys can't excrete more than about 0.8–1.0 l/hour; extreme exertion with over-hydration; no electrolyte replacement; stress, compulsive drinking.

Prevention:

- In moderation: max. **0.5–0.8 litres/hour**, spread evenly.
- Replace electrolytes in heat and on long missions.
- Listen to thirst – don't drink compulsively "in advance", but do drink regularly.

> ⚠ Water is a strategic resource: both waste and over-drinking are harmful.

4.6 Thickening of the blood

Haemoconcentration depends not only on water intake but also on fluid loss and electrolyte balance.

Causes: large fluid loss (sweat, breath, urine, diarrhoea); heat + exertion without salt and water; severe dehydration (thrombosis risk); too much salt or protein.

The right balance: water + electrolytes together; under exertion 0.5–0.8 l/hour, in heat salt tablets or electrolyte powder.

4.7 Finding water sources SUPPLEMENTED

Always look for the cleanest source first: **spring > running water > still water**.

- Valley bottoms, green vegetation at the foot of rocks, reeds, sedges, willow and alder indicate water.
- Game trails going downhill often lead to water; swarming insects, birds at dawn and dusk.
- Dig on the outer bend of a dry stream bed for seeping water.
- Rainwater: collected on a tarp or poncho – the cleanest natural source.
- **Dew:** wipe grass with a cloth at dawn and wring it out.
- Snow and ice: **melt it, don't eat it!** – it chills your body and dries your mouth. Ice gives more water for less fuel.
- **Transpiration bag:** a clear bag over a living, leafy branch in the sun – condensation collects in a few hours.
- **Solar still:** a pit with a container in the middle, plastic sheet over it, weighed down with a stone in the centre.

⚠ Never drink seawater, urine or alcohol – they increase dehydration.

⚠ Avoid water near dead animals, rubbish, industrial sites or farmland, and water that is foamy, oily, smelly or lifeless.

4.8 Filtering and clarifying SUPPLEMENTED

Filtering removes cloudiness but **does not disinfect** – always purify afterwards.

1. Settling: let it stand, then gently pour off the top.
2. Improvised filter (bottle, sock, trouser leg), layers from top to bottom: grass/cloth → fine sand → charcoal (from the fire, not ash) → gravel.
3. Pouring through cloth (scarf, T-shirt) – against larger debris.

✔ Charcoal reduces smell and taste but does not kill pathogens.

4.9 Purifying water SUPPLEMENTED

1. **Boiling – the most reliable:** a rolling boil for at least **1 minute**; above 2000 m, **3 minutes**. Let it cool covered.
2. **Chlorine dioxide tablets** (e.g. Micropur): against bacteria, viruses and Cryptosporidium too. About 30 minutes; with cold or cloudy water up to **4 hours**.
3. **Iodine tablets / tincture:** about 30 minutes; longer in cold water. Not during pregnancy, with thyroid disease, or long-term.
4. **Chlorine bleach (unscented, 5–6%):** 2 drops/litre of clear, 4 drops/litre of cloudy water, 30 minutes; a slight chlorine smell should remain.
5. **Filter bottle, microfilter:** against bacteria and protozoa; against viruses only combined with chemicals.
6. **UV (SODIS):** clear PET bottle, all day in strong sun – only as a last resort.

⚠ After treating with tablets, loosen the cap and turn the bottle over so the thread is disinfected too.

⚠ Add vitamin C drink powder only after the treatment time – it reduces the effect.

4.10 Storing and rationing SUPPLEMENTED

- In a clean, closable, labelled container, in the shade; never mix treated and untreated water.
- Don't hoard to the last drop: water is better in your body than in your bottle – but ration it.
- Reduce water loss: rest in the shade during the hottest hours, move at night, don't smoke, talk little, breathe through your nose.

5 Evasion

Part II, Lesson 1 — based on the course outline, extended from public sources

5.1 Purpose of evasion SUPPLEMENTED

Evasion is the set of procedures by which the isolated person avoids detection and contact by hostile or potentially hostile forces and gets back to friendly forces or is extracted.

> ✔ Captivity is worse than evasion! Every decision aims to stay unseen and return.

- First step after isolation: leave the landing/accident site immediately – that's where they look first.
- Take only the necessary gear; hide or destroy the rest.
- Move to a hide, then S-T-O-P: stop, think, observe, plan.

5.2 Hiding and alternative destinations SUPPLEMENTED

- Choose a hide by the **"LOW" principle**: low silhouette, odd (irregular) shape, wild/blends into the terrain – not the obvious bush or building.
- Primary destination: designated extraction site / friendly forces. Alternatives: secondary extraction point, neutral territory, border, allied forces.
- Always have a plan B: escape route, rally point.

5.3 Camouflage

Camouflage aims to make the isolated person, their gear and their shelter unrecognisable to the enemy's senses and sensors. Camouflage is an active task – it must be maintained continuously.

5.4 Types of observation SUPPLEMENTED

The five senses: the enemy sees, hears, smells (tobacco, soap, cooking, fire), feels (tracks) and tastes – guard against all of them.

Direct observation:

- Physical signs: footprints, broken branches, trampled grass, discarded litter, traces of fire.
- Locals: dogs, shepherds, hunters, farmers – they are the best "sensors".

Indirect observation: aerial and drone reconnaissance ("the view from above" – regular shapes, shadows and paths stand out), thermal and night-vision devices, radio direction finding.

5.5 Recognition factors SUPPLEMENTED

What the human eye spots immediately – eliminate these:

1. **Movement** – the biggest giveaway; move slowly and deliberately, freeze when observed.
2. **Position** – not on the skyline, not on prominent points, not in the obvious spot.
3. **Shape** – break up the outline of the human body, helmet and weapon.
4. **Shadow** – your own shadow gives you away; use nature's deep shadows.
5. **Texture** – smooth surfaces look lighter and shine; rough surfaces look darker.
6. **Colour** – mottled colouring that matches the surroundings.

7. **Tone** – differences in brightness; the glowing patch of face and hands.
8. **Shine** – watch, glasses, metal, optics, sweaty skin – cover them, use matt surfaces.

> ✔ Memory aid: **movement – shape – shadow – shine – silhouette – spacing**.

5.6 Principles of camouflage SUPPLEMENTED

- **Site:** fits the surroundings, inconspicuous, several exits.
- **Camouflage discipline:** leave no tracks or litter; avoid paths.
- **Light discipline:** no light at all at night (torch, match, screen, cigarette) – visible from far away.
- **Against night vision and thermal:** hide body heat behind soil, rocks, dense vegetation or an emergency blanket (used correctly); not in open ground.
- **Noise discipline:** secure your gear, whisper, hand signals, don't snap branches.
- **Concealed shelter:** irregular shape, local material, hidden from aerial view too.

5.7 Personal camouflage SUPPLEMENTED

- **Shape:** break up your outline with local vegetation (on clothing, helmet, pack); change it as the surroundings change.
- **Background:** always match the background – dark clothing stands out against a light background.
- **Techniques:** face paint: raised areas (forehead, nose, cheekbones, chin) darker, hollows lighter; hands and neck too.
- **Blending:** before moving, check how you fit the surroundings; move slowly and irregularly.

5.8 Concealment in temperate zones SUPPLEMENTED

- On clothing: fresh, local vegetation – not all one kind, not too much.
- Natural materials: mud, soil, charcoal for the face; leaves, moss.
- For shelters: irregular shape, local material, no straight edges; covered from above as well.
- **Fire while hiding:** only if essential; small smokeless fire (dry hardwood), Dakota fire hole, under the canopy; by day smoke and by night light give you away.

5.9 Getting food during evasion SUPPLEMENTED

- Concealment comes first – foraging is a risk.
- Animals: traps and snares (quiet, passive); hunting or firing a weapon makes noise – last resort only.
- Plants: only species you know are edible; if in doubt, don't eat.
- In inhabited areas: avoid taking anything that will be missed; abandoned gardens and orchards – carefully.
- Preparation: clean and cook raw animal food and water.

5.10 Security and movement SUPPLEMENTED

Precautions: stop, look, listen and smell regularly (STOP–LOOK–LISTEN–SMELL). Before moving, plan the route and the next hide.

Observing terrain: even without binoculars: sector by sector, near to far; watch the skyline and the shadows.

Main reasons evaders get caught: risk-taking from impatience, fatigue, hunger or thirst; using roads, bridges and villages; tracks; asking unreliable people for help.

Movement:

- Use poor light (dawn, dusk, night).
- Route: avoid roads, paths, ridgelines and open ground; move below the crest, through dips, along the edge of vegetation.
- Hide tracks: hard ground, stones, water's edge; step in already trampled spots.
- Reduce the chance of discovery: short bounds, frequent stops, changes of direction.

Ways of moving: stalking walk, running (only if detected), crawling (low, high, side).

5.11 Rally points SUPPLEMENTED

A pre-designated, easily recognisable terrain feature where the group regroups after being scattered.

- Recognisable but not conspicuous; with a set waiting time and an onward plan.
- At the rally point: security, head count, treat the wounded, assess, plan.
- Obstacles: rivers, roads, fences, towns, minefields – cross them only when justified and after reconnaissance.

5.12 Evasion aids SUPPLEMENTED

- **Personal survival kit (PSK):** fire starter, water purification, signalling devices, compass, knife, cord, first aid, food, map – always on your body.
- **Maps and charts:** evasion chart (with blood chit), code tables.
- **Bilingual phrase card (“Pointee-Talkee”):** picture-and-text sheet in the local language for communicating by pointing.
- **Assisted evasion:** an organised, vetted network of helpers.
- **Unexpected help:** a local helper – high risk, only when justified.
- **Contact with locals:** only as a last resort, with someone alone, after observing; don't reveal more than necessary.

6 Land navigation

Part II, Lesson 2 — based on the course outline, extended from public sources

6.1 Introduction SUPPLEMENTED

Orientation is the basis of evasion and extraction: without accurate positioning neither reporting nor link-up is possible. Tools: map, compass, terrain features, and the last resort – the sun and stars.

6.2 Reading a grid reference SUPPLEMENTED

Military maps use the MGRS/UTM grid: grid lines are 1 km apart (on a 1:50,000 map).

1. Find the bottom-left corner of the square containing the point.
2. **"Along the corridor, then up the stairs"**: read the easting (vertical line) first, then the northing (horizontal line).
3. Refine with tenths (protractor): 6 digits = 100 m, 8 digits = 10 m accuracy.

> ✔ Example: 045-245 → easting 04.5 / northing 24.5 grid units.

6.3 Measuring distance on the map SUPPLEMENTED

- Scale 1:50,000 → 1 cm on the map = **500 m** on the ground; 1:25,000 → 1 cm = 250 m.
- Straight line: with a ruler / the compass's 1:50,000 scale.
- Winding road: strip of paper, thread, or add up segments.
- On the ground: **pace count** – measure in advance how many paces make 100 m (average 60–70); more uphill, at night, loaded.

6.4 Orienting the map SUPPLEMENTED

On an oriented map, directions match those on the ground.

1. **With a compass:** lay the compass edge along a north grid line and turn the map with the compass until magnetic north is on the mark (taking declination into account).
2. **By terrain features:** turn the map to the ground using a straight road, railway or distant identifiable point.

6.5 Protractor (coordinate scale) SUPPLEMENTED

A transparent sheet with degree/mil scales and grid-reference scales.

- Measuring a bearing on the map: centre on the start point, 0° on grid north, read the direction to the target.
- Convert the **grid** azimuth to **magnetic** using the map's declination diagram before using it with a compass.

6.6 Measuring a bearing with the Cammenga 3H compass SUPPLEMENTED

Parts of the Cammenga lensatic compass: cover with sighting wire, lens eyepiece, floating dial, rotating bezel (click ring), thumb loop, 1:50,000 scale. The 3H model's tritium illumination is visible at night without charging.

Reading a bearing (compass-to-cheek method):

1. Open the cover to 90°, the eyepiece to about 45° – the dial must float freely.
2. Thumb through the loop, index finger along the side; support it from below with the other hand.
3. Raise it to eye level, thumb against the cheekbone.
4. Look through the eyepiece: line the target up with the sighting wire.
5. Read the bearing on the dial under the sighting wire.

Setting a bearing: the clicking bezel: 1 click = **3°** (or 50 mils); 360° = 120 clicks. It can be set in the dark by counting clicks.

> ⚠ Near metal objects, vehicles, power lines or weapons the compass errs – step away (at least 1–2 m from a weapon, 10+ m from a vehicle).

6.7 Measuring an azimuth in the field SUPPLEMENTED

Measuring:

1. Choose the target.
2. Sight it along the wire with the compass-to-cheek hold.
3. Wait for the dial to settle, read it.
4. Repeat for accuracy.

Following a bearing:

1. Turn the dial (or click the bezel) to the desired azimuth.
2. With the centre hold, turn your body to align the needle with the luminous mark.
3. Pick a feature on that line, walk to it, repeat.

> ✔ At night: the two luminous dots and the needle lining up show the direction.

6.8 Modified resection SUPPLEMENTED

If you're on a known line (road, river, ridge):

1. Measure the azimuth to a known feature identified on the map.
2. Calculate the back azimuth (±180°).
3. Draw the back azimuth from the feature on the map – where it crosses the known line is where you are.

6.9 Resection SUPPLEMENTED

If your position is unknown:

1. Orient the map, identify **two (better three)** clearly visible features also on the map, ideally 60–120° apart.
2. Measure each azimuth, calculate the back azimuths (±180°).
3. Draw the lines from the features – the intersection (centre of the triangle) is your position.

6.10 Navigating without tools SUPPLEMENTED

- **Shadow stick:** mark the tip of a stick's shadow, again 15 minutes later – the first mark is west, the second east.

- **Watch:** in the northern hemisphere point the hour hand at the sun; the bisector between it and 12 (1 in summer time) points south.
- **North Star:** extend the distance between the Big Dipper's two pointer stars five times – north.

> ⚠ Moss, tree rings and anthills are unreliable direction indicators.

7 Extraction - link-up procedures

Part III, Lesson 1 — based on the course outline, extended from public sources

7.1 Why JPR matters SUPPLEMENTED

Personnel Recovery (PR) is the sum of military, diplomatic and civil efforts to recover and reintegrate isolated personnel. Successful recovery boosts morale, prevents the enemy from using a captured person for propaganda or intelligence, and preserves trained personnel.

7.2 Basic terms SUPPLEMENTED

- **ISOP (IP):** isolated personnel.
- **JPR:** Joint Personnel Recovery.
- **CSAR:** combat search and rescue – one method of the "Recover" task, not a synonym for PR.
- **JPRC:** Joint Personnel Recovery Centre; **PRCC:** Personnel Recovery Coordination Cell.
- **ISOPREP:** pre-filled identification data sheet (question–answer, photo, fingerprints) for authentication.
- **EPA:** Evasion Plan of Action – the planned route and intentions.

7.3 European and national aspects SUPPLEMENTED

- **EDA (European Defence Agency):** from 2013 launched personnel recovery planner and controller courses (PRCPC, later JPRSC) to improve interoperability between member states.
- **EPRC (European Personnel Recovery Centre):** established on 8 July 2015, based at Poggio Renatico air base in Italy. An intergovernmental military organisation harmonising PR rules and training.

> ✔ Hungary is active: the Air Command and Control Centre in Veszprém regularly hosts EDA JPR staff courses, with Hungarian instructors.

7.4 Operational elements: JPRC, PRCC SUPPLEMENTED

- **JPRC:** directs and coordinates all PR activity in the theatre at operational command level.
- **PRCC:** coordinates recovery at component/national level and liaises with the JPRC.
- **Links:** in NATO, EU and UN operations national PR elements plug into the multinational structure; the aim is common procedures and fast help.

7.5 Phases of the JPR system SUPPLEMENTED

1. **Preparation:** training (SERE), equipment, ISOPREP, planning.
2. **Planning:** PR plans, EPA, communication procedures.
3. **Execution:** the five execution tasks.
4. **Assessment:** processing lessons learned.

7.6 Education and training SUPPLEMENTED

Personnel recovery rests on three pillars: commanders and staffs, recovery forces, and the **isolated person** themselves. The isolated person's preparedness (SERE, ISOPREP, EPA, equipment) decisively affects the success of recovery.

7.7 The five PR execution tasks SUPPLEMENTED

1. **Report:** begins with recognising the isolating event; must be fast and accurate.
2. **Locate:** finding and authenticating the person; recovery forces only launch with an accurate position and positive ID.
3. **Support:** supporting the person and their family – two-way communication, resupply drops, suppressing enemy forces.
4. **Recover:** coordinated action of commanders, recovery forces and the isolated person.
5. **Reintegrate:** medical check, debriefing, return to duty and family – physical and mental health come first.

> ✔ The isolated person takes part too: making radio contact helps the Report and Locate tasks.

7.8 International commitments SUPPLEMENTED

Soldiers serving in NATO and EU operations (e.g. KFOR) work in environments where a PR system is available – SERE preparedness fits into it.

7.9 Purpose of link-up SUPPLEMENTED

Link-up aims for the isolated person and the recovery/friendly force to meet **safely and with mutual authentication** – without fratricide or falling into an enemy trap.

7.10 Steps before physical contact SUPPLEMENTED

1. Reach the designated extraction/rally site and observe the surroundings.
2. Check in according to the given procedure (radio, signal) – only on instruction.
3. Prepare signalling devices (mirror, flare, strobe, IR signal, panel).
4. Equipment: weapon on safe, nothing suspicious in your hands.
5. Follow the recovery force's instructions exactly.

> ⚠ To a recovery force an unknown armed person is a threat – keep your hands visible.

7.11 Authentication SUPPLEMENTED

- **ISOPREP questions:** pre-set personal questions only the isolated person can answer.
- **Number, letter, colour codes:** according to the daily/operational codes.
- **Sound, hand and light signals:** in a pre-agreed sequence.
- **Checks:** the recovery force may put the isolated person on the ground and search them – this is a security procedure, not hostility.

7.12 Real example - Libya 2011 SUPPLEMENTED

On 21 March 2011, during Operation Odyssey Dawn, a US F-15E Strike Eagle crashed near Benghazi due to a technical failure. The two crew members ejected but landed far apart.

- The pilot was picked up by a Marine **TRAP** package launched from USS Kearsarge (2 MV-22 Ospreys, 2 CH-53Es, 2 AV-8B Harriers) within about 90 minutes of the report; time on the ground was under 90 seconds.

- The weapon systems officer was found by local rebels and later handed over to US forces.

> ✔ Lesson: fast reporting, accurate positioning and authentication, a recovery force on standby – and a prepared isolated person.

8 What is expected of the person to be recovered

Part III, Lesson 2 — based on the course outline, extended from public sources

8.1 Staying alive SUPPLEMENTED

The isolated person's needs in order – the SERE priorities:

1. **Personal protection:** first aid, shelter, clothing, fire – against weather and enemy.
2. **Sustenance – water and food:** water comes before food.
3. **Health – physical and psychological:** treating injuries, hygiene, sleep, morale, the will to survive.
4. **Travel:** only with good reason and a plan – either wait for rescue on the spot, or evade/exfiltrate.

> ✔ Memory aid: the **"rule of threes"** – 3 minutes without air, 3 hours unprotected in extreme weather, 3 days without water, 3 weeks without food.

8.2 Keeping your dignity SUPPLEMENTED

- **Captivity – worse than evasion!** Seize every opportunity to avoid capture.
- **Resistance:** in captivity the isolated person follows the code of conduct: gives only the details required by international law (Geneva Conventions) – name, rank, date of birth, service number; does not cooperate with the enemy, makes no propaganda statements.
- **Escape:** if an opportunity arises and the risk is acceptable, it is a duty to try to escape – the first hours after capture are the most favourable.

> ✔ Faith, comradeship, routine and discipline: the foundations of surviving captivity mentally.

8.3 Preparing before deployment SUPPLEMENTED

- Fill in and update your ISOPREP; know the EPA (evasion plan).
- Comms plan: frequencies, call signs, codes, check-in windows.
- Check the survival kit and signalling devices on your body.
- Know the area: map, extraction sites, rally points, danger areas.

8.4 Reporting the isolating event SUPPLEMENTED

- Report as soon as you can: who, where, when, condition, intentions, threat.
- With a radio: a short, coded transmission in the given window from a concealed spot – then move.
- Without a radio: use signalling devices at the right time at the extraction site.

8.5 Informing the recovery forces SUPPLEMENTED

- Exact position (grid), how to authenticate, injuries, enemy forces, the terrain and the landing site.
- Listen and follow instructions; don't make unnecessary transmissions.

8.6 The recovery site SUPPLEMENTED

- Concealed close to the designated site, with enough open ground for the recovery asset.
- An obstacle-free landing site (for helicopters: no trees, wires or loose objects); wind direction.
- Signal only when the recovery force approaches / on instruction.

8.7 Physical contact in brief SUPPLEMENTED

1. Signal as instructed.
2. Weapon on safe, hands visible, don't run toward the recovery asset.
3. Authenticate yourself (ISOPREP questions).
4. Accept being searched, follow the recovery force's instructions.
5. When boarding, follow the recovery force's commands – head down, away from the rotor.

B1 Advanced fire, shelter and weather

BRAVO · based on ATP 3-50.21 and AFH 10-644

B1.1 Lighting a fire without matches or a lighter SUPPLEMENTED

At level B one lighter is not enough: have at least two other ways to make fire, and practise them in dry and wet weather.

- **Ferrocerium rod:** strike down with the knife spine or a scraper, right above the tinder; works in rain too.
- **Magnifier, glasses, clear bottle of water:** focus sunlight on one point and dark tinder will ignite.
- **Battery + steel wool:** the two terminals of a 9 V battery, or two AA cells in series, instantly heat fine steel wool.
- **Bow drill:** dry softwood (lime, poplar, willow), a notch in the fireboard, steady and accelerating strokes – takes a lot of practice.

Tinder in wet weather: the inside of standing deadwood, feather sticks (thin shavings left attached to the stick), birch bark (oily, burns even when wet), resinous pine stumps.

> ⚠ One fire starter is no fire starter. Keep several methods in your survival kit, waterproofed, on your body.

B1.2 Fire types by task SUPPLEMENTED

- **Teepee:** catches quickly; for cooking and signalling.
- **Log cabin (criss-crossed sticks):** makes plenty of coals and holds heat; for cooking.
- **Star (logs radiating, burning in the middle):** burns slowly and economically; push the logs in during the night.
- **Dakota fire hole:** two connected holes (fire pit + air tunnel); low flame, little smoke and light – a concealed fire.
- **Reflector wall:** logs or stones behind the fire; directs heat toward the shelter.

> ✔ In hostile territory: small, dry hardwood, under the canopy; smoke gives you away by day, light by night. Don't light a fire unless you must.

B1.3 Advanced shelters SUPPLEMENTED

- **Lean-to + reflector wall:** back to the wind, fire in front, wall behind the fire – the warmest simple solution.
- **Debris hut:** ridgepole and ribs covered with 60–90 cm of dry leaves; warm even without fire, only slightly bigger than your body.
- **Tarp A-frame:** cord between two trees, tarp stretched over it; quick, best in rain.
- **Snow trench and snow cave:** snow is a good insulator. Entrance lower than the sleeping platform (cold air drains out), always a ventilation hole, keep the digging tool inside.
- **Hammock, raised bed:** over wet ground or ground full of creatures.

> ⚠ A fire or stove burning in a snow cave or tent: carbon monoxide poisoning. Always ventilate!

B1.4 Cold and heat injuries SUPPLEMENTED

Hypothermia signs: violent shivering, then clumsiness, stumbling, mumbling, confusion, and finally shivering stops – this is already life-threatening.

- Action: sheltered dry place, insulation below and above, dry clothes, a warm (not hot) sweet drink if conscious, body-to-body warmth.

Frostbite: white, waxy, numb skin.

- Don't rub it, don't warm it by the fire. Only rewarm with body heat if it can't refreeze.

Heat exhaustion: pale, clammy skin, weakness, headache → shade, rest, water + salt.

Heat stroke: hot red skin, confusion, unconsciousness → **cool immediately** (water, wet cloth, wind); life-threatening.

Trench foot: feet wet and cold for days. Prevention: dry socks for the night, daily foot care.

B2 Water and Food in the Field

BRAVO · based on ATP 3-50.21 and AFH 10-644

B2.1 Advanced water procurement SUPPLEMENTED

- **Transpiration bag:** a clear bag tied over a living, sunny, leafy branch; the plant can give one or two cups of water a day. Never on a poisonous plant.
- **Solar still:** a hole about 1 m wide, a container in the middle, plastic sheet over it with a small stone in the centre. Little water for a lot of work – best for purifying salty or dirty water.
- **Dew:** wipe the grass with a cloth at dawn, then wring it out.
- **Plants:** birch and maple sap in spring; a cut wild grapevine drips water.
- **Snow and ice:** melt it, don't eat it (it chills and dehydrates you). Put a little water in the bottom of the pot so it doesn't scorch.

⚠ Purify all natural water: boiling, tablets, or filter + disinfection. Filtering alone does not kill pathogens.

B2.2 Edible and poisonous plants SUPPLEMENTED

Only eat what you **recognise for certain**. Plant food gives little energy, but provides vitamins and fibre.

- **Safe and easy to recognise (Central Europe):** nettle (cooked), dandelion, plantain, sorrel, wild raspberry, blackberry, rose hips, hazelnuts, acorns (leached in several changes of water), inner bark of pine, pine-needle tea.
- **Avoid:** plants with milky sap, a bitter-almond smell, umbrella-shaped flower heads (hemlock!), leaves in threes, and white or yellow berries.
- **Never eat mushrooms** in the field – the deadly species can be mistaken for edible ones.
- **Yew:** every part is deadly poisonous (including the seed of the red berry) – never make tea from it.

Edibility test (only in an emergency, if there's nothing else for days): one plant part, first on the skin, then the lip, the tongue, then a small bite, and wait 8 hours. Slow, and it doesn't protect against every poison.

B2.3 Animal food SUPPLEMENTED

- **Insects:** the easiest protein to get (grubs, grasshoppers, ant pupae). Always boil or roast them. Avoid hairy, brightly coloured, stinging and bad-smelling ones.
- **Fish:** improvised hook (needle, thorn), line, bait; a fish trap in shallow water; always cook fish thoroughly.
- **Small game with snares:** on game trails, at narrow points, in front of burrows. Many snares, checked at least once a day.

⚠ In peacetime, hunting, trapping and fishing require a licence. These methods are for genuine survival situations only.

✓ The body can last weeks without food, but only days without water. Getting food must not cost more energy than it gives.

B2.4 Preparing and storing food SUPPLEMENTED

- Clean animals away from the water source and your camp; bury the remains (smell, predators, tracks).
- Cook everything thoroughly: the best protection against parasites and bacteria.
- **Preserving:** cut into thin strips and smoke them, or dry them in the sun and wind.
- Boiling keeps most nutrients – drink the broth too.
- In hostile territory the smell and smoke of cooking give you away: eat cold food there if you can.

B3 Advanced navigation

BRAVO · based on ATP 3-50.21 and FM 3-25.26

B3.1 Navigating by the stars SUPPLEMENTED

- **North Star (northern hemisphere):** find the Big Dipper and measure five times the distance between the two pointer stars at the end of the "pan" – there is the North Star, and north is on the horizon below it.
- **Cassiopeia:** the "W"-shaped constellation opposite the Big Dipper, on the other side of the North Star – useful when the Dipper is hidden.
- **Orion's Belt:** rises in the east, sets in the west; the line of three stars runs roughly east–west.

Star-movement method: line up two sticks on a star and check after 15–20 minutes which way it moved. Up → you're facing east, down → west, right → south, left → north.

B3.2 Sun and watch SUPPLEMENTED

- **Shadow stick:** push a stick into the ground, mark the tip of the shadow; mark it again after 15 minutes. The first mark is west, the second east; the line between them runs east–west.
- **Analogue watch (northern hemisphere):** point the hour hand at the sun; halfway between the hour hand and 12 is south. During summer time use 1 instead of 12.
- **At local noon** the sun is due south, and the shadow is shortest and points north.

⚠ Moss, tree rings and anthills are not reliable direction indicators – don't rely on them.

B3.3 Dead reckoning and distance SUPPLEMENTED

Dead reckoning: from a known point you travel a measured distance on a known bearing (azimuth), constantly working out where you are.

- **Pace count:** measure your own value over 100 m beforehand (flat, slope, dense vegetation separately). Mark every 100 m (a knot in a cord, a pebble in your pocket).
- **Bypassing an obstacle:** turn out at a right angle counting paces, continue on the original bearing, then come back to the line with the same number of paces.
- **Aiming off:** if you're looking for a point on a linear feature (road, stream), deliberately aim to one side of it – then you'll know which way to turn when you reach it.
- At night and in fog, distances seem longer and your pace is shorter – allow for it.

B3.4 Reading the terrain SUPPLEMENTED

- **Handrail:** a linear feature (stream, ridge, forest edge, power line) you can follow with certainty, without constantly checking the compass.
- **Catching feature:** a landmark beyond your objective; if you reach it, you know you've gone too far.
- **Ridge, saddle, valley, spur:** the shapes of contour lines – practise matching the map to the real terrain.
- If you're unsure: stop (S-T-O-P), go back to the last certain point, don't guess.

✓ In the field, water running downhill (stream → river) usually leads to settlements and roads – but when evading, those are exactly the dangerous places.

B4 Evasion in the field, signalling, link-up

BRAVO · based on FM 21-76-1 and ATP 3-50.21

B4.1 Planning movement in hostile territory SUPPLEMENTED

- **Cycle:** observe → move → observe. Short bounds, frequent halts, and always know where you'll hide if you must.
- **Time of day:** move before dawn, at dusk and at night; rest in a hide by day.
- **Route:** dense vegetation, low ground, shaded slopes. Avoid roads, trails, ridgelines, open ground and settlements.
- **Crossing a road:** at a bend or where the view is shortest; observe for a long time first, then cross at right angles, quickly, one at a time.
- **Crossing a stream:** at the widest, shallowest point, with footwear on, angled against the current; keep spare clothes dry.

> ⚠ Impatience and fatigue are the main reasons evaders get caught. Better slow but unseen.

B4.2 Tracks and how to hide them SUPPLEMENTED

Anything you leave can be found: footprints, broken twigs, flattened grass, dew brushed off, overturned stones, litter, smell, faeces.

- Step on hard ground, rock, roots and leaf litter; avoid soft, muddy ground.
- Don't break branches; bend them and let them spring back.
- Bury rubbish and faeces, and leave your hide as you found it.
- Walking in water is only worth it for short stretches – it chills you, and the exit point gives you away.
- Don't come back the same way; vary your direction but keep your general heading.

B4.3 Defeating observation SUPPLEMENTED

- **Drones, aerial reconnaissance:** if you hear one, **freeze** – movement is what stands out most. Stay under the canopy, in shade, among irregular shapes.
- **Thermal imaging:** body heat glows against a cold background. Hide behind soil, rock or dense vegetation; open ground offers no good hiding place.
- **Night vision:** the smallest light is visible from far away: phone, watch, cigarette.
- **Search dogs:** a dog follows scent, not sight. Time and distance are your best protection – keep moving, don't panic, don't try to fight.
- **Electronics:** a switched-on phone reveals your position. Turn it on only in the planned time window.

B4.4 Improvised signalling and link-up SUPPLEMENTED

- **Signal mirror:** make a V with two fingers, frame the target (aircraft) in the V, and walk the light spot between your fingers. In sunshine it can be seen tens of kilometres away.
- **Ground signals** (at least 3 m long, high contrast): **V** need assistance, **X** need medical help, **N** no, **Y** yes, → travelling this way.
- **Radio / phone:** only in the time window of the communications plan, with a short message (who, where, condition).

- **Linking up with rescuers:** stay hidden until you've identified them; weapon on safe, hands visible, don't run toward them; follow instructions exactly, answer the authentication questions (ISOPREP).

> ✔ Signalling cuts both ways: rescuers and the enemy both see it. Assess the threat first.

C1 Risk of capture and how to prevent it

CHARLIE · based on AFH 10-644 and public NATO sources

C1.1 Who is level C for? SUPPLEMENTED

Level C is for **high-risk** personnel: those whose task, post or rank makes isolation and capture likely, and who carry information of special value to the enemy.

- Aircrew, special operations forces, reconnaissance, long-range patrols.
- Liaison and monitoring teams, personnel serving abroad, senior personnel.
- In peacetime too: the risk of hostage-taking or arbitrary detention abroad.

> ⚠ The core of level C is practical training under controlled conditions with experienced instructors. This chapter gives the theory only; it does not replace the course.

C1.2 Preparation before you go SUPPLEMENTED

- **ISOPREP:** fill it in and keep it up to date; rescuers use these personal questions to authenticate you.
- **Evasion plan of action (EPA):** route, rally points, signals, time windows – in your head, not on paper.
- **Personal items:** only what the task needs. Don't take letters, photos, notes or unnecessary documents; your phone should hold nothing that exposes you, your family or your unit.
- **Survival kit on your body:** if you lose your pack or vehicle, this stays with you.
- **Fitness and health:** captivity and evasion test the body too.

C1.3 The moment of capture SUPPLEMENTED

Capture is one of the most dangerous phases: the captors are tense and the situation is chaotic.

- If resistance is hopeless, **don't provoke**: hands visible, slow movements, follow instructions.
- Stay calm and observe: how many there are, where they take you, what you see and hear – you may need this to escape later.
- Ask for care for yourself and your wounded comrades.
- Don't start conversations, don't speculate aloud, don't make threats.

> ✓ How you behave in the first hours matters: a calm, disciplined prisoner attracts less violence.

C1.4 Escape in the early phase SUPPLEMENTED

According to public doctrine, the best chance to escape is often **soon after capture**: friendly forces are still close, the captors are disorganised, and you are still at your strongest.

- Weigh it up: how good the chance is, your health, where you would go, and what failure would mean – for you and your comrades.
- If you go: at once, decisively, then apply the rules of evasion (B4).
- If not: observe, gather information, save your strength for a later opportunity.

C2 Rights and conduct of a prisoner of war

CHARLIE · based on the Third Geneva Convention and AFH 10-644

C2.1 Protection of the Third Geneva Convention SUPPLEMENTED

A soldier captured in an international armed conflict is a **prisoner of war**, protected by the Third Geneva Convention (1949).

- Entitled to humane treatment; torture, intimidation and public humiliation are prohibited.
- Entitled to food, water, clothing, accommodation and medical care.
- May correspond with family and be visited by the International Committee of the Red Cross (ICRC).
- **Under questioning, only four items must be given:** surname and first names, rank, date of birth, service number. No other information may be forced out of a prisoner (Article 17).

⚠ Not every adversary respects the Convention. Knowing your rights still gives you strength, and it is the basis for holding them accountable later.

C2.2 You remain a soldier in captivity SUPPLEMENTED

- Captivity does not end military discipline: the senior prisoner leads, the others support.
- Help your comrades; don't accept favours that harm them.
- Don't make statements or sign anything that could be used against your country, the allies or your comrades.
- Keep your human dignity: washing, order, bearing – these are part of survival.

C2.3 Interrogation - general principles SUPPLEMENTED

The aim of interrogation is to gain information and often propaganda.

- Stick to the four items and deflect everything else politely and calmly.
- Methods vary: kindness, promises, fatigue, isolation, misleading “information” about your comrades. Expect them and don't be surprised.
- Don't believe that “the others have already told us everything”.
- If you slip, don't give up: go back to your starting position at the next question.

✓ Specific resistance techniques are practised with level C course instructors under controlled conditions.

C2.4 Peacetime detention and hostage situations SUPPLEMENTED

During peace operations, missions or travel abroad, a detained person is **not a prisoner of war**, so the situation is different.

- The first minutes are the most dangerous: don't resist armed people, follow instructions.
- Stay human in your captors' eyes: name, family, calm voice – don't be submissive, but don't provoke either.
- Eat, drink and sleep when you can; keep your routine.

- During a rescue: **lie down**, hands visible, don't run, don't grab anything, follow the rescuers' instructions – at first they treat everyone as a suspect.

C3 Resilience and survival in captivity

CHARLIE · based on AFH 10-644

C3.1 Stress and the will to survive SUPPLEMENTED

The greatest enemy in captivity is often not the captor but hopelessness.

- Fear, anger, guilt and sleep problems are normal – normal reactions to an extraordinary situation.
- **Faith and purpose:** family, comrades, coming home, faith or your values – hold on to whatever gives you strength.
- Remember that your country and the allies are working to free you.

> ✔ According to returned prisoners of war, the key to survival was hope and small daily goals.

C3.2 Daily routine SUPPLEMENTED

- **Exercise:** even in a small space: squats, push-ups, stretching – maintenance for body and mind.
- **Mental exercise:** learn, plan, recall memories, build a house in your head, solve problems.
- **Hygiene:** wash as much as you can, keep your space in order.
- **Eating and drinking:** eat what you are given, even if it's bad – you need your strength.
- **Sense of time:** count the days, build a daily schedule.

C3.3 Contact with fellow prisoners SUPPLEMENTED

- Comrades keep each other alive: look out for each other, share your strength.
- The senior prisoner leads the group, and the chain of command works in captivity too.
- Captors try to prevent communication. Vietnam-era POWs, for example, used the publicly known **tap code**: a 5×5 letter grid, tapping first the row, then the column number.
- Don't spread rumours, don't accuse your comrades – division helps the captor.

C3.4 Health in captivity SUPPLEMENTED

- Report illness or injury and ask for treatment – it's your right.
- Keep wounds as clean as you can; watch for signs of infection (redness, swelling, fever, foul smell).
- Water and food may be scarce: pace yourself, rest when you can.
- Watch your comrades' state of mind too: whoever withdraws, stops eating or gives up needs you.



C4 Escape, extraction and reintegration

CHARLIE · based on AFH 10-644 and public NATO sources

C4.1 The decision to escape SUPPLEMENTED

- **When?** The earlier, the better the chance – but only if the situation is favourable. Plan a later escape thoroughly.
- **Weigh up:** physical condition, knowledge of the terrain, where friendly forces are, what happens to your comrades if you leave.
- **Preparation:** gather information (guard changes, terrain, weather), save strength and, if possible, food.
- **Afterwards:** apply the rules of evasion at once (B4) – move at night, hide your tracks, make contact according to the plan.

> ⚠ A failed escape can have serious consequences. The decision is your responsibility, with the senior prisoner's knowledge if possible.

C4.2 Extraction and rescue SUPPLEMENTED

- Rescue forces work fast and decisively and treat everyone as dangerous until identified.
- **Don't run toward them.** Lie down or stay where you are, hands visible, don't grab anything.
- Follow instructions to the letter; accept being searched and restrained.
- Authentication: answer the ISOPREP questions, tell them where other comrades are.

> ✔ At the moment of rescue, discipline saves lives – yours too.

C4.3 Reintegration SUPPLEMENTED

Release is not the end of the story. The reintegration process in NATO countries has several phases:

1. **Medical care and rest:** assessing your physical condition, treatment.
2. **Debriefing:** what you know may help those still in captivity and future training.
3. **Psychological support:** a specialist helps you process what happened.
4. **Gradual return** to family and duty.

- Sleep problems, irritability, guilt or withdrawal lasting for months are normal. They can be treated – ask for and accept help.

C4.4 What you can do now SUPPLEMENTED

- Fill in and update your ISOPREP; know your communications and evasion plan by heart.
- Keep up your fitness and survival knowledge (levels A and B).
- Put your papers in order, tell your family who to contact if something happens.
- Practise stress management: breathing, routine, small goals.
- Apply for an official level B or C course if your post justifies it.

N1 Nuclear situations and doctrine

EXTRA · based on public sources

N1.1 What is a nuclear doctrine? SUPPLEMENTED

A **nuclear doctrine** is a state's public set of rules on when, why and on whose authority it may use nuclear weapons. Its main aim is **deterrence**: the other side should not dare to attack, because the response would cause unacceptable damage.

- **NATO:** the Alliance states that as long as nuclear weapons exist, it will remain a nuclear alliance. Its deterrence rests on the nuclear forces of the USA, the UK and France, and on nuclear sharing.
- **Russia:** the new doctrine approved on 19 November 2024 replaces the earlier "threat to the very existence of the state" with a **critical threat to sovereignty or territorial integrity** as a possible reason.
- The same document treats an attack by a non-nuclear state supported by a nuclear state as a **joint attack**, and extends deterrence to states that provide territory for an attack.

> **A doctrine is a political signal. As soldiers and civilians, our job is not to guess, but to know what to do if trouble comes anyway.**

> Doctrines change – always check the current texts and official information.

N1.2 Effects of a nuclear explosion SUPPLEMENTED

1. **Flash:** blinding light that can cause temporary blindness for up to a minute.
2. **Blast wave:** can cause fatal injuries and building damage for kilometres. It arrives seconds after the flash, later at greater distances.
3. **Heat and fire:** burns and fires over a radius of kilometres.
4. **Initial radiation:** at the moment of the explosion, mainly close by.
5. **Electromagnetic pulse (EMP):** can destroy electronics; causes temporary disruption further away.
6. **Radioactive fallout:** contaminated dust and debris carried high up falls back down. It is dangerous to **people outdoors**, and usually reaches more distant areas **more than 15 minutes** after the explosion.

> The best protection against fallout is thick walls and earth: the more material between you and the open sky, the lower the radiation.

N1.3 At the moment of the flash SUPPLEMENTED

1. **Don't look at the light.** Turn away and close your eyes.
2. **Lie face down** behind cover (a wall, ditch or hole), so as little skin as possible is exposed.
3. Cover your head and neck with your arms. Don't touch your eyes, nose or mouth.
4. **Stay down until the blast wave has passed** – this can take seconds, and a return wind may follow.
5. **In a vehicle:** stop safely and duck below the level of the seats.
6. If you are warned in advance: go into the nearest building immediately and keep away from windows.

N1.4 Get inside, stay inside, stay tuned SUPPLEMENTED

> ★ GET INSIDE · STAY INSIDE · STAY TUNED

Get inside: after the blast wave you have **about 10 minutes or more** to find shelter before fallout arrives. Vehicles, tents and lightweight houses do not protect you.

- Best place: a **basement**, or the **middle** of a large brick or concrete building, away from outer walls and the roof.
- In a tall building, the inner part of the middle floors is better than the top floor or the windowed part of the ground floor.

Stay inside: for at least **24 hours** unless the authorities say otherwise. Don't set off to fetch your family – everyone should stay where they are and meet later. Keep pets inside too.

Stay tuned: a battery or wind-up radio works even when the mobile network, TV and internet are down. The authorities will say when and where it is safe to go out.

> ✔ Close doors and windows, and switch off ventilation and air conditioning that draws air from outside.

N1.5 The 7:10 rule SUPPLEMENTED

Radiation from fallout drops quickly. A simple rule of thumb: **every sevenfold increase in time cuts the radiation to a tenth.**

- Example: if it is 400 units **2 hours** after the explosion, it is about 40 after **14 hours**, and about 4 two days later.
- That is why **the first hours are the most dangerous** – shelter matters most then.

> ⚠ A rule of thumb, not a measurement. A radiation meter is more accurate if you have one and know how to use it – base your decision on official instructions.

N1.6 Decontamination: if you were outside SUPPLEMENTED

1. Before or straight after entering the building, **take off your outer layer of clothing**, put it in a bag, tie it and put it away from people.
2. **Shower** with lukewarm water and soap. If you can't: wipe exposed skin and hair with a clean, damp cloth.
3. **Don't scrub** the skin hard, and don't use hand sanitiser or disinfectant wipes to remove fallout.
4. Brush pets gently, then wash them with soap and water.
5. Put on clean clothes that were stored indoors.

N1.7 Food, water, health SUPPLEMENTED

- **Packaged** food and drink, and food and drink **stored indoors**, are safe. Wipe the packaging before opening.
- Don't eat or drink anything that was **left uncovered outdoors**.
- Go for medical care when the authorities say so; in a public shelter, tell staff at once if you are ill or injured.

- **Shelter supplies** for at least 24 hours, preferably 3 days: water, food, medicines, a battery or wind-up radio, torch, batteries, soap, wet wipes.

N1.8 Nuclear power plant accident in Hungary SUPPLEMENTED

In Hungary the **National Nuclear Emergency Response System (ONER)** monitors radiation, runs the alert system and decides on protecting the population.

Alerts: sirens, radio, television, local loudspeakers. If you hear a siren: go inside and turn on the radio or TV.

Sheltering at home:

1. Come inside, keep the family together, bring pets in too.
2. Close doors, windows and vents; seal gaps if needed.
3. Turn on the radio or TV and follow the instructions.
4. Get out the information leaflet you received earlier and prepare for further instructions.

Iodine tablets: only take them if the authorities **order it** – they only protect against radioactive iodine, not other radiation. The authorities will say where they are handed out and how to take them.

> ⚠ Don't take iodine tablets "just in case" on your own decision.

Evacuation: only when instructed. Take: personal documents, medicines, mobile phone with charger, warm clothes. Leave by your own car, or if you have none, by the buses sent to your settlement.

> ✔ Official radiation data: the disaster management radiation monitoring site and the gamma dose rate map of the Hungarian Meteorological Service.

N1.9 As a soldier in the field SUPPLEMENTED

- **Cover first:** the best protection from fallout is a place dug into the ground and covered from above (cellar, bunker, covered trench). Open ground and vehicles do not protect you.
- **Don't move in the first hours** if you have cover – by the 7:10 rule, waiting greatly reduces your radiation dose.
- **Follow CBRN procedures** and your commander's orders: protective equipment, decontamination, marking and avoiding contaminated areas.
- Keep water and food clean: only use sealed packs that have been wiped.
- **Report** any contamination you find and any casualties.

> ⚠ This is a practice aid – it does not replace the current CBRN regulations and training.

Part III

Practice exam questions

Questions chapter by chapter, with an answer key at the end. A test is passed from 70%.

1. chapter · The SERE training system

1. What does the acronym SERE stand for?

A) Survival, Reconnaissance, Hiding, Withdrawal
B) Survival, Evasion, Resistance, Extraction
C) Security, Control, Alert, Supply
D) Search, Rescue, Supply, Evacuation

2. Which risk category does SERE level A belong to?

A) Special
B) High
C) Medium
D) Low

3. What does ISOP stand for?

A) Personal survival kit
B) International rescue procedure
C) Extraction helicopter
D) Isolated personnel – separated from friendly forces

4. What is the aim of Resistance?

A) Avoiding the enemy in the field
B) Directing the rescue forces
C) To reduce the effectiveness of interrogation tactics in captivity
D) Armed combat against the captors

5. What is JPR?

A) Joint patrol system
B) Field medical care
C) Joint Personnel Recovery
D) Prisoner exchange procedure

6. How long are SERE levels B and C valid?

A) 1 year
B) Indefinitely
C) 3 years
D) 5 years

7. By how much can a refresher extend level B/C?

A) It cannot be extended
B) By 3 years
C) By 1 year
D) By 5 years

8. Which NATO standard is the national SERE system built on?

A) STANAG 1059
B) STANAG 4569
C) STANAG 2116
D) STANAG 7196

9. What is the pass mark for chapter tests?

A) 90%
B) 60%
C) 50%
D) 70%

10. What is Extraction?

A) Interrogating a prisoner
B) Removing equipment
C) Treating the casualty on site
D) Recovering the isolated person from hostile territory

2. chapter · Fire making

1. What are the three elements of the fire triangle?

A) Wood, matches, wind
B) Heat, smoke, flame
C) Fuel, heat, oxygen
D) Tinder, kindling, water

2. What is the correct order of fuel when lighting a fire?

A) Kindling → tinder → fuel wood
B) Tinder → kindling → fuel wood
C) Fuel wood → kindling → tinder
D) The order doesn't matter

3. How much fuel should you gather, by rule of thumb?

A) Exactly as much as needed
B) Tinder alone is enough
C) Three times your estimate
D) Half, then top up as you go

4. Why must bamboo and elder be split?

A) Its smoke is poisonous
B) Trapped air inside can make them explode
C) Otherwise it won't burn
D) It's too heavy

5. Why not build a fire over pine roots?

A) Pine doesn't burn
B) There will be too much smoke
C) The roots absorb the heat
D) Resinous roots can start a forest fire days later

6. Why are wet riverbank stones dangerous around a fire?

A) They give off toxic gas
B) They can burst from the heat
C) They put the fire out
D) They're not dangerous

7. In cold weather, which injury is more common around a fire?

A) Burns
B) Dehydration
C) Frostbite
D) Fracture

8. On what ground do you always need a platform under the fire?

A) Sandy ground
B) Grassy meadow
C) Humus-rich, peaty ground
D) Rocky ground

9. Which fire gives little light and smoke, good for concealment?

A) Long (Finnish) fire
B) Log cabin fire
C) Dakota fire hole
D) Teepee fire

10. What should you do if you must leave the fire unattended?

A) Bank it safely or put it out
B) Water the ground around it
C) Nothing, it goes out by itself
D) Add more wood

3. chapter · Building shelters

1. When should you start building a shelter?

A) When it's already dark
B) Only when it starts raining
C) At least 1–2 hours before dark
D) In the morning, after waking up

2. Why avoid the valley bottom for the night?

A) Cold air flows down there at night
B) You can't make a fire there
C) It's too warm there
D) There's no wood there

3. What are “widowmakers”?

A) Hidden gullies
B) Dead, half-fallen trees and branches that can fall
C) Predators
D) Poisonous berries

4. What pitch is recommended for a non-waterproof roof?

A) 10–20°
B) Under 30°
C) 90°
D) 45–60°

5. At what angle should the shelter opening be to the prevailing wind?

A) at 90°
B) It doesn't matter
C) Back to the wind, 180°
D) Facing the wind

6. What is the ideal distance between fire and shelter?

A) Approx. 90–100 cm
B) 10–20 cm
C) 5 metres
D) 20 metres

7. Why is a ground pad (ground insulation) important?

A) The ground draws the most heat – without it you'll freeze even at 0 °C

B) Not important if you have a sleeping bag

C) Only for comfort

D) It protects against insects

8. What does a mat's R-value indicate?

A) Its weight

B) Its waterproofness

C) Its size

D) Its insulating ability – the higher, the better

9. In a snow shelter, what does it mean if you can't see your breath?

A) There's not enough oxygen

B) It's too cold

C) Everything's fine

D) It's too warm inside, cool it down

10. In the cold, what should you do with your clothing before building?

A) Nothing

B) Put on every layer

C) Take off some layers so you don't sweat

D) Soak your jacket

4. chapter · Finding and purifying water

1. Roughly how long can a person survive without water?

A) 3 months

B) 3 weeks

C) 3 hours

D) 3 days

2. What is the recommended fluid intake under physical exertion?

A) 0.1 litres/hour

B) Only when you're thirsty

C) 0.5–0.8 litres/hour

D) 2–3 litres/hour

3. What causes hyponatraemia (water intoxication)?

A) Too much water in a short time, without electrolytes

B) Drinking salt water

C) Too much salt

D) Too little water

4. What is the ratio for a survival electrolyte drink?

A) 1 pinch of salt / 10 litres of water

B) No salt needed

C) ½ teaspoon salt / 1 litre water

D) 5 tablespoons of salt / 1 litre of water

5. Which is the most reliable water purification method?

A) Filtering through cloth

B) Settling

C) Boiling

D) Leaving it in the sun

6. How long should you boil water above 2000 m?

A) 1 hour

B) Until it boils

C) 10 seconds

D) At least 3 minutes

7. What does an improvised sand-charcoal filter NOT do?

A) It doesn't clear cloudiness

B) It doesn't disinfect (doesn't kill pathogens)

C) It does everything

D) It doesn't improve the taste

8. How should you use snow for drinking water?

A) Eat it raw

B) Eat only the ice

C) Melt it, don't eat it

D) Not at all

9. Which drink increases dehydration?

A) Boiled stream water

B) Purified rainwater

C) Electrolyte water

D) Seawater

10. When can you add vitamin C drink powder to tablet-treated water?

A) Right away, together with the tablet
B) After the treatment time is up
C) Before
D) Never

8. chapter · Evasion

1. What's the first thing to do after becoming isolated?

A) Shout for help
B) Wait in one place
C) Leave the landing / separation site
D) Make a fire

2. What is the most revealing recognition factor?

A) Tone
B) Texture
C) Movement
D) Colour

3. Why avoid the skyline?

A) There's no signal there
B) Because it's cold there
C) No need to avoid it
D) Your silhouette stands out

4. What is light discipline about?

A) Use red light only
B) Fire only at night
C) Torch pointed only at the ground
D) No light at all at night (torch, screen, cigarette)

5. How should you paint your face for camouflage?

A) Evenly black
B) Light colour everywhere
C) Raised areas darker, hollows lighter
D) Only the forehead

6. When is it best to move during evasion?

A) Any time
B) Only by day
C) At noon
D) At dawn, dusk, night

7. What is a rally point?

A) The helicopter landing site
B) The enemy position
C) The field kitchen
D) A pre-designated point where the group regroups

8. What is a “Pointee-Talkee”?

A) A bilingual point-to phrase card with pictures
B) A type of compass
C) A radio call sign
D) Signal mirror

9. What is one of the main reasons evaders get captured?

A) Moving at night
B) Taking risks out of impatience and fatigue
C) Moving too slowly
D) Too much camouflage

10. Where should the personal survival kit (PSK) be?

A) In the vehicle
B) At the base
C) On your body
D) At the bottom of the pack

6. chapter · Land navigation

1. On a 1:50,000 map, how many metres on the ground is 1 cm?

A) 50 m
B) 5 km
C) 1 km
D) 500 m

2. In what order do you read a grid reference?

A) Down, then left
B) It doesn't matter
C) Right, then up (easting, then northing)
D) Up, then right

3. How many degrees is one click of the Cammenga compass bezel?

A) 3°
B) 5°
C) 10°
D) 1°

4. What is a back azimuth?

A) Twice the azimuth
B) The azimuth ± 180°
C) The azimuth ± 90°
D) Always 0°

5. At least how many landmarks do you need for a resection?

A) No landmarks needed
B) One
C) Five
D) Two (better three)

6. When do you use a modified resection?

A) When you're on a known linear feature (road, river)
B) Only with GPS
C) At night
D) When your position is completely unknown

7. What deflects a compass?

A) Metal, vehicles, weapons, power lines
B) Wind
C) Cold
D) Sunlight

8. How do you find the North Star?

A) Extend the distance between the Big Dipper's two pointer stars five times
B) Next to the Moon
C) Always on the horizon
D) The brightest star

9. What do the two marks of a shadow stick show?

A) First mark: west, second: east
B) First: north, second: south
C) Only the time
D) Nothing

10. Is moss on a tree a reliable direction indicator?

A) No
B) Yes, it's always on the south side
C) Yes, it's always on the north side
D) Only in winter

7. chapter · Extraction - link-up procedures

1. What is the correct order of the five PR execution tasks?

A) Locate, Recover, Report, Reintegrate, Support
B) Support, Report, Recover, Locate, Reintegrate
C) Report, Locate, Support, Recover, Reintegrate
D) Recover, Report, Locate, Support, Reintegrate

2. What is CSAR?

A) Medical evacuation
B) Combat search and rescue – one method of recovery
C) A synonym for PR
D) A type of compass

3. What is ISOPREP?

A) A pre-filled identification data sheet
B) A radio frequency
C) A survival kit
D) A map type

4. Where is the European Personnel Recovery Centre (EPRC) based?

A) Poggio Renatico (Italy)
B) Berlin
C) Veszprém
D) Brussels

5. Which Hungarian location hosts EDA JPR staff courses?

A) Szolnok
B) Pápa
C) Tata
D) Veszprém

6. What should you do when the recovery force arrives?

A) Shout and wave your weapon
B) Weapon on safe, hands visible, follow instructions
C) Run toward the helicopter
D) Hide from them

7. What is the purpose of link-up?

A) Deceiving the enemy
B) Getting food
C) A safe, mutually authenticated meeting
D) Radio transmission

8. Why accept being searched during recovery?

A) Because you're a prisoner
B) Only for the wounded
C) It's a security procedure, not hostility
D) You don't have to

9. Libya 2011: what aircraft did the US crew go down in?

A) B-2
B) F-15E Strike Eagle
C) F-16
D) Tornado

10. Libya 2011: how long was the TRAP force on the ground picking up the pilot?

A) About 2 hours
B) All night
C) Less than 90 seconds
D) About 30 minutes

8. chapter · What is expected of the person to be recovered

1. What is the “rule of threes”?

A) 3 days without sleep, 3 weeks without water, 3 months without food
B) 3 hours of walking a day
C) 3 fires, 3 shelters, 3 signals
D) 3 minutes without air, 3 hours without protection, 3 days without water, 3 weeks without food

2. According to the SERE A course, what is the first need?

A) Travel
B) Personal protection
C) Food
D) Escape

3. Under the Geneva Conventions, which details must you give in captivity?

A) Your unit's location and strength
B) The operation plan
C) Name, rank, date of birth, service number
D) Nothing

4. When is escape most likely to succeed?

A) In the first hours after capture
B) Only at night, after a year
C) Never
D) After months

5. What is an EPA?

A) Body armour
B) First aid kit
C) Radio code
D) Evasion plan of action

6. When should you signal the recovery force?

A) On instruction or when the rescuer approaches
B) Never
C) Only by day
D) Continuously

7. What should you do after a radio transmission?

A) Make a fire
B) Switch it off for good
C) Change position
D) Stay there and wait

8. What should the recovery site be like?

A) Clear of obstacles, with a concealed waiting spot nearby
B) Under power lines
C) The middle of a dense forest
D) A village square

9. What do you do when boarding a helicopter?

A) Run toward the tail rotor
B) Stand up and wave
C) Head down, away from the rotor, as instructed
D) Climb in on your own

10. What is the basis of surviving captivity mentally?

A) Isolating yourself from others
B) A propaganda statement
C) Cooperating with the enemy
D) Faith, comradeship, routine, discipline

Bravo 1. chapter · Advanced fire, shelter and weather

1. Why do you need several fire-starting methods at level B?

A) Because a fire must always be concealed
B) Because matches are banned
C) Because a lighter is too heavy
D) Because one tool can be lost or get wet

2. Which tinder burns even when wet?

A) Wet moss
B) Fresh pine needles
C) Green willow leaves
D) Birch bark

3. Which fire gives a low flame and little smoke and light?

A) Signal fire with green leaves
B) Log cabin
C) Dakota fire hole
D) Teepee

4. How thick should the leaf layer of a debris hut be?

A) 1–2 cm
B) No leaves needed
C) 60–90 cm
D) 5–10 cm

5. Where should the entrance of a snow cave be?

A) On the top
B) Lower than the sleeping platform
C) Level with the sleeping platform
D) No entrance needed

6. What is a dangerous sign of hypothermia?

A) Feeling slightly cold
B) Shivering stops and confusion sets in
C) Goose bumps
D) Feeling hungry

7. What should you NOT do with frostbite?

A) Get to a sheltered place
B) Keep it dry
C) Rewarm with body heat if it can't refreeze
D) Rub it or warm it by the fire

8. What is the first aid for heat stroke?

A) Immediate cooling
B) Hard exercise
C) A warm blanket
D) Lots of salt without water

Bravo 2. chapter · Water and food in the field

1. What is a solar still best for?

A) Cooking food
B) Purifying salty or dirty water
C) Signalling
D) Large amounts of drinking water

2. How do you get water from snow?

A) Keep it in your pocket
B) Eat it in big mouthfuls
C) Mix it with salt
D) Melt it, don't eat it

3. Is filtering enough to make natural water safe?

A) Yes, always
B) No, it must also be disinfected
C) Only in winter
D) Except for spring water

4. What should you do with mushrooms in the field?

A) Eat only small ones
B) Eat only colourful ones
C) Never eat them
D) Eat them raw

5. Which plant is deadly poisonous in every part?

A) Dandelion
B) Yew
C) Nettle
D) Plantain

6. How should you eat insects?

A) Always boiled or roasted
B) Only hairy ones
C) Only brightly coloured ones
D) Raw, alive

7. May you set snares in peacetime?

A) Only in forests
B) Only at weekends
C) Yes, anywhere
D) No, it needs a licence; only in a real survival situation

8. Why not clean an animal next to the water source?

A) Because of smell, predators, tracks and contamination
B) There is no such rule
C) Because it's cold there
D) Because there's no light

Bravo 3. chapter · Advanced navigation

1. How do you find the North Star?

A) It's always the brightest star
B) Five times the distance between the Big Dipper's two pointer stars
C) In the middle of Orion's Belt
D) To the right of the Moon

2. Star-movement method: the star moves up. Which way are you facing?

A) South
B) North
C) West
D) East

3. Shadow-stick method: what does the first mark show?

A) East
B) West
C) North
D) South

4. Analogue-watch method (northern hemisphere): what points south?

A) The number 6
B) Halfway between the hour hand and 12
C) The minute hand
D) The second hand

5. Is moss on a tree a reliable direction indicator?

A) Yes, it always grows on the south side
B) No
C) Yes, it always grows on the north side
D) Only in winter

6. What is dead reckoning?

A) Using GPS
B) Counting stars
C) Going back to the start
D) Tracking your position by measured distance on a known bearing from a known point

7. What does aiming off mean?

A) Deliberately aiming to one side of the objective on a linear feature
B) Deceiving the enemy
C) Rotating the compass
D) A random change of direction

8. What is a “handrail” in navigation?

A) The edge of the map
B) A linear feature you follow
C) The compass lanyard
D) A stair rail

Bravo 4. chapter · Evasion in the field, signalling, link-up

1. What is the basic movement cycle in hostile territory?

A) Move only by day
B) Move without stopping
C) Observe – move – observe
D) Run – rest – run

2. How should you cross a road?

A) In a group, slowly
B) On the straightest, most visible stretch
C) After observing, at right angles, quickly, one at a time
D) Walking along the road

3. What is one of the main reasons evaders get caught?

A) Moving at night
B) Too much rest
C) Impatience and fatigue
D) Moving slowly

4. What should you do if you hear a drone?

A) Light a fire
B) Freeze and stay under cover
C) Run into the open
D) Wave

5. What protects against thermal imaging?

A) Moving at night in the open
B) Nothing
C) Dark clothing in the open
D) Cover behind soil, rock or dense vegetation

6. Which ground signal means: need medical help?

A) N
B) Y
C) V
D) X

7. How should you approach rescuers?

A) Running and shouting
B) Weapon in hand
C) From behind, silently
D) Stay hidden until you identify them; hands visible, don't run

8. What does a switched-on phone reveal?

A) Your position
B) Nothing
C) Only the temperature
D) Only your name

Charlie 1. chapter · Risk of capture and how to prevent it

1. Who is SERE level C for?

A) Civilians only
B) Every recruit
C) Doctors only
D) Personnel at high risk of capture

2. What is ISOPREP for?

A) Rescuers use it to authenticate you
B) Weapon records
C) Food requests
D) Requesting leave

3. What should you NOT take on a dangerous task?

A) A survival kit
B) Letters, photos, unnecessary documents
C) Water
D) A compass

4. How should you behave at the moment of capture if resistance is hopeless?

A) Shout and threaten
B) Don't provoke, hands visible, follow instructions
C) Run at any cost
D) Attack at once

5. Why is it important to observe during capture?

A) So you don't get bored
B) The information may be needed for escape later
C) So you can complain
D) It isn't important

6. When is the best chance to escape often?

A) Never
B) In the early phase after capture
C) Years later
D) Only during interrogation

7. What should you weigh up before escaping?

A) Only the weather
B) Chance, health, destination, consequences of failure
C) Nothing, go at once
D) Only your own comfort

8. Does this chapter replace the level C course?

A) Yes, completely
B) Yes, if you read it twice
C) Only for officers
D) No, it is only the theoretical basis

Charlie 2. chapter · Rights and conduct of a prisoner of war

1. Which convention protects prisoners of war?

A) Hague Convention on cultural property
B) Ottawa Convention
C) Schengen Agreement
D) Third Geneva Convention

2. What information must a prisoner of war give?

A) Name, rank, date of birth, service number
B) Their family's address
C) Location of their unit
D) The battle plan

3. Who leads a group of prisoners?

A) The youngest
B) Nobody
C) The captor
D) The senior prisoner

4. What should you do if asked to sign a statement?

A) Sign it to get better treatment
B) Sign them all
C) Correct it and sign
D) Don't sign anything that can be used against your country or comrades

5. What should you expect under interrogation?

A) Only shouting
B) Nothing
C) Immediate release
D) Alternating kindness, promises, fatigue, deception

6. What should you do if you slipped up in interrogation?

A) Stop eating
B) Attack the interrogator
C) Don't give up, return to your starting position
D) Tell everything

7. Is a soldier taken hostage in peacetime a prisoner of war?

A) No, the legal status is different
B) Only in uniform
C) Only at weekends
D) Yes, always

8. What should you do during a rescue?

A) Lie down, hands visible, don't run
B) Hide from the rescuers
C) Run toward the rescuers
D) Grab a weapon

Charlie 3. chapter · Resilience and survival in captivity

1. What is one of the greatest enemies in captivity?

A) Lack of boredom
B) Too much sleep
C) Hopelessness
D) Cold food

2. Are fear and sleep problems normal in captivity?

A) Only for officers
B) Yes, a normal reaction to an extraordinary situation
C) They never occur
D) No, a sign of weakness

3. Why is exercise important even in a small space?

A) It isn't important
B) Maintenance for body and mind
C) So the guards see
D) To get tired

4. What should you do with the food the captors give you?

A) Eat only half
B) Eat it, you need your strength
C) Give it to the guards
D) Refuse it in protest

5. What was a well-known way Vietnam POWs communicated?

A) Mobile phone
B) Tap code with a 5×5 letter grid
C) Smoke signals
D) Radio

6. How do you strengthen the group in captivity?

A) Accuse the weaker ones
B) Accept favours at others' expense
C) Look out for comrades, don't spread rumours
D) Isolate yourself

7. What is a sign of infection in a wound?

A) A scar without itching
B) Fast healing
C) Redness, swelling, fever, foul smell
D) Cold skin

8. What should you do if a comrade stops eating and withdraws?

A) Take their food
B) Report them to the guards
C) Engage with them, support them
D) Leave them alone for good

Charlie 4. chapter · Escape, extraction and reintegration

1. When is the chance of escape greater?

A) Only years later
B) Never
C) The earlier, if the situation is favourable
D) Only by day

2. What do you do after a successful escape?

A) Move at once by the rules of evasion
B) Light a fire and signal
C) Go back alone for the others
D) Rest next to the camp

3. How should you behave when rescuers break in?

A) Run toward them
B) Hide
C) Grab the captor's weapon
D) Lie down, hands visible, follow instructions

4. Why do rescuers treat you as a suspect?

A) Because they don't believe you
B) Until identified, they treat everyone as dangerous
C) They don't
D) As a punishment

5. What is the first step of reintegration?

A) Medical care and rest
B) Returning to duty at once
C) A press conference
D) Work without leave

6. Why is debriefing important?

A) So you can be punished
B) It isn't important
C) It's just paperwork
D) It may help those still in captivity and training

7. Are sleep problems lasting months after release normal?

A) Yes, and treatable – ask for help
B) No, it's weakness
C) Only for civilians
D) They never occur

8. What can you do now to prepare?

A) Update ISOPREP, fitness, knowledge, papers in order
B) Just wait
C) Keep everything secret from your family
D) Nothing

☢ Extra 1. chapter · Nuclear situations and doctrine

1. What is the main aim of a nuclear doctrine?

A) Keeping weapons secret
B) Rapid territorial gain
C) Protecting nuclear power plants
D) Deterrence

2. What should you do outdoors at the moment of a nuclear flash?

A) Lie face down behind cover
B) Stand up to see better
C) Look to see where the explosion was
D) Start running to the nearest car

3. Roughly how long do you have after the blast wave to find shelter before fallout?

A) Exactly 2 hours
B) A whole day
C) A few seconds
D) About 10 minutes or more

4. Where is the best shelter from radioactive fallout?

A) In a tent
B) On the top floor, next to a window
C) In a car with the windows closed
D) In a basement or the middle of a large concrete building

5. How long should you stay inside at least, unless the authorities say otherwise?

A) Until the rain stops
B) 30 minutes
C) 2 hours
D) 24 hours

6. What does the 7:10 rule say?

A) It is dangerous within 7 km for 10 minutes
B) After 7 hours radiation is 10 times higher
C) After 10 hours radiation is 7 times lower
D) Every sevenfold increase in time cuts radiation to a tenth

7. What is the first decontamination step if you were outside?

A) Wash the clothes and put them back on
B) Rub hand sanitiser on your skin
C) Scrub your skin hard with a brush
D) Take off the outer layer of clothing and bag it

8. Which food is safe after fallout?

A) Rainwater
B) Packaged or indoor-stored food with wiped packaging
C) An uncovered bowl of food left outdoors
D) Fruit picked in the garden

9. When should you take iodine tablets?

A) When you hear a siren
B) Every morning as prevention
C) When you have a headache
D) Only if the authorities order it

10. What do iodine tablets protect against?

A) The blast wave
B) Only radioactive iodine
C) All radiation
D) Burns

11. What still works when the mobile network and internet are down?

A) A smartphone news feed
B) A Wi-Fi smart TV
C) A video call
D) A battery or wind-up radio

12. What should you do during an evacuation?

A) Set off on foot at once to fetch your family

B) Stay at home despite all instructions

C) Leave only when instructed, with documents and medicines

D) Leave your medicines behind to travel light

Answer key

1. chapter · The SERE training system

1. B	**2.** D	**3.** D	**4.** C	**5.** C	**6.** D
7. B	**8.** D	**9.** D	**10.** D		

2. chapter · Fire making

1. C	**2.** B	**3.** C	**4.** B	**5.** D	**6.** B
7. A	**8.** C	**9.** C	**10.** A		

3. chapter · Building shelters

1. C	**2.** A	**3.** B	**4.** D	**5.** A	**6.** A
7. A	**8.** D	**9.** D	**10.** C		

4. chapter · Finding and purifying water

1. D	**2.** C	**3.** A	**4.** C	**5.** C	**6.** D
7. B	**8.** C	**9.** D	**10.** B		

5. chapter · Evasion

1. C	**2.** C	**3.** D	**4.** D	**5.** C	**6.** D
7. D	**8.** A	**9.** B	**10.** C		

6. chapter · Land navigation

1. D	**2.** C	**3.** A	**4.** B	**5.** D	**6.** A
7. A	**8.** A	**9.** A	**10.** A		

7. chapter · Extraction - link-up procedures

1. C	**2.** B	**3.** A	**4.** A	**5.** D	**6.** B
7. C	**8.** C	**9.** B	**10.** C		

8. chapter · What is expected of the person to be recovered

1. D	**2.** B	**3.** C	**4.** A	**5.** D	**6.** A
7. C	**8.** A	**9.** C	**10.** D		

Bravo 1. chapter · Advanced fire, shelter and weather

1. D	**2.** D	**3.** C	**4.** C	**5.** B	**6.** B
7. D	**8.** A				

Bravo 2. chapter · Water and food in the field

1. B	**2.** D	**3.** B	**4.** C	**5.** B	**6.** A
7. D	**8.** A				

Bravo 3. chapter · Advanced navigation

1. B	**2.** D	**3.** B	**4.** B	**5.** B	**6.** D
7. A	**8.** B				

Bravo 4. chapter · Evasion in the field, signalling, link-up

1. C **2.** C **3.** C **4.** B **5.** D **6.** D
7. D **8.** A

Charlie 1. chapter · Risk of capture and how to prevent it

1. D **2.** A **3.** B **4.** B **5.** B **6.** B
7. B **8.** D

Charlie 2. chapter · Rights and conduct of a prisoner of war

1. D **2.** A **3.** D **4.** D **5.** D **6.** C
7. A **8.** A

Charlie 3. chapter · Resilience and survival in captivity

1. C **2.** B **3.** B **4.** B **5.** B **6.** C
7. C **8.** C

Charlie 4. chapter · Escape, extraction and reintegration

1. C **2.** A **3.** D **4.** B **5.** A **6.** D
7. A **8.** A

Extra 1. chapter · Nuclear situations and doctrine

1. D **2.** A **3.** D **4.** D **5.** D **6.** D
7. D **8.** B **9.** D **10.** B **11.** D **12.** C

Notes, important numbers

Emergency number: 112 – say WHO you are, WHERE you are, WHAT happened.

Name:

Important phone numbers: